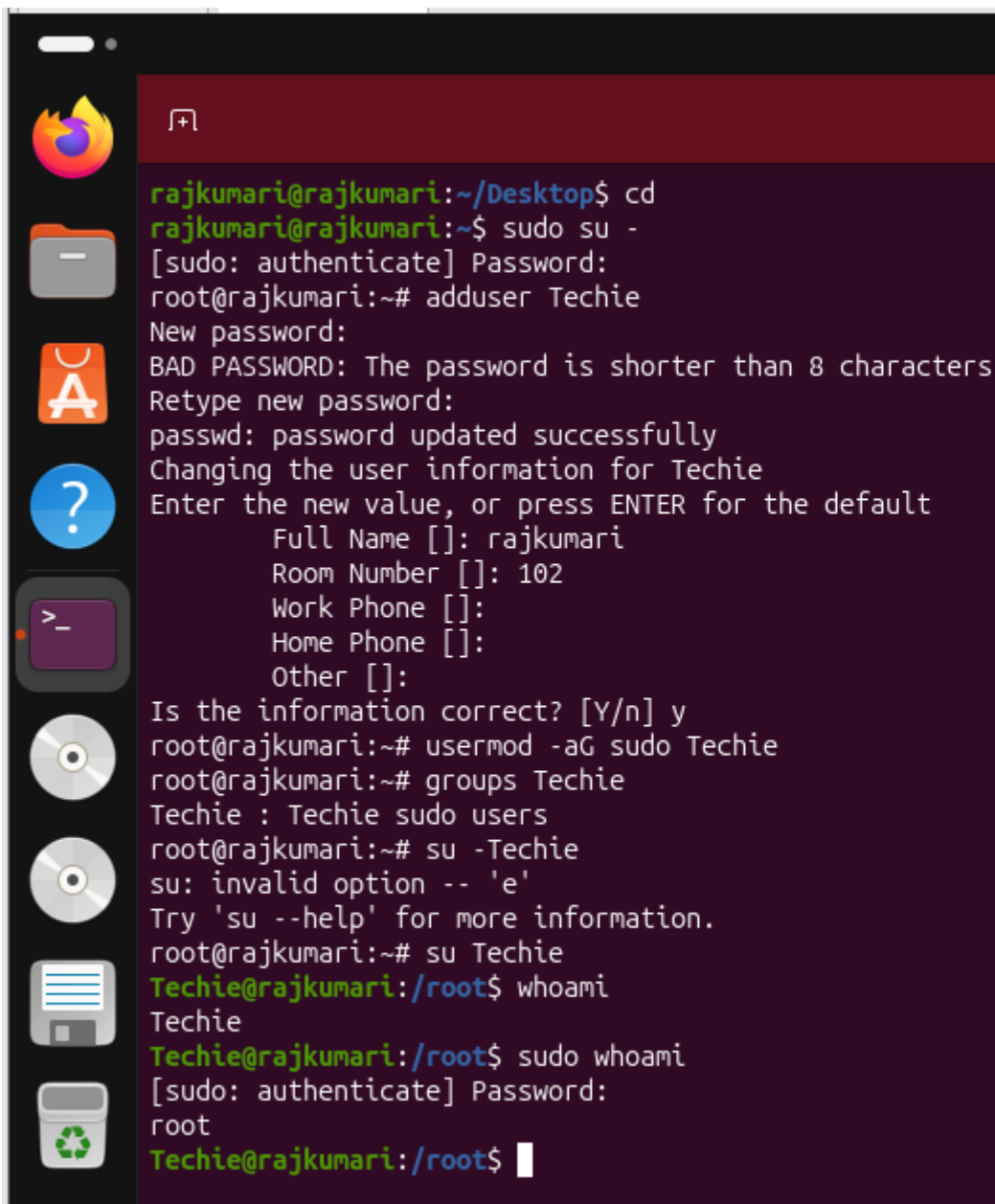


1. Create user with name Techie and provide sudo access to user

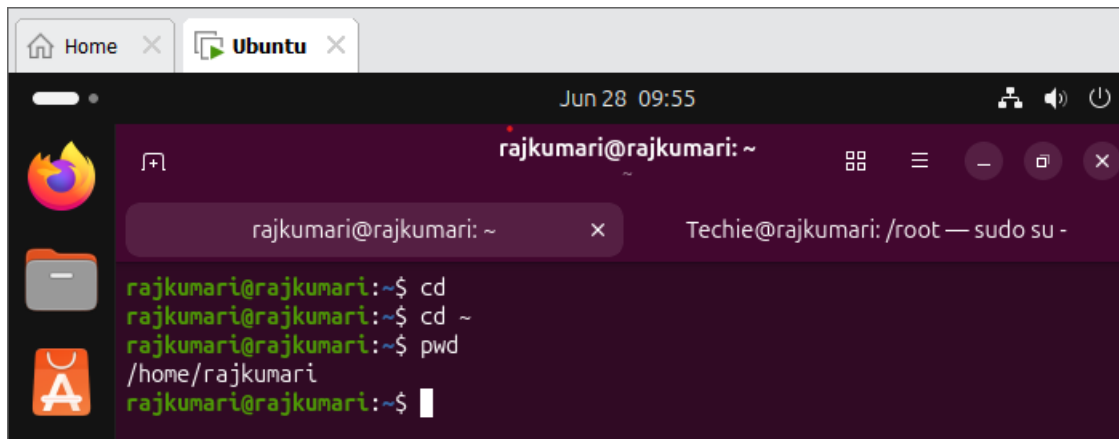


```
rajkumari@rajkumari:~/Desktop$ cd
rajkumari@rajkumari:~$ sudo su -
[sudo: authenticate] Password:
root@rajkumari:~# adduser Techie
New password:
BAD PASSWORD: The password is shorter than 8 characters
Retype new password:
passwd: password updated successfully
Changing the user information for Techie
Enter the new value, or press ENTER for the default
    Full Name []: rajkumari
    Room Number []: 102
    Work Phone []:
    Home Phone []:
    Other []:
Is the information correct? [Y/n] y
root@rajkumari:~# usermod -aG sudo Techie
root@rajkumari:~# groups Techie
Techie : Techie sudo users
root@rajkumari:~# su -Techie
su: invalid option -- 'e'
Try 'su --help' for more information.
root@rajkumari:~# su Techie
Techie@rajkumari:/root$ whoami
Techie
Techie@rajkumari:/root$ sudo whoami
[sudo: authenticate] Password:
root
Techie@rajkumari:/root$
```

2. How to Navigate to the home directory.

Objective: Navigate to the current user's home directory.

Purpose: Quickly return to your personal working directory from anywhere in the file system.



A terminal window titled 'rajkumari@rajkumari: ~' with a dark purple background. The window shows the following commands and output:

```
rajkumari@rajkumari:~$ cd
rajkumari@rajkumari:~$ cd ~
rajkumari@rajkumari:~$ pwd
/home/rajkumari
rajkumari@rajkumari:~$
```

3. Create a new directory

Objective: Create a new directory (folder) in the current location.

Purpose: Organize files by storing them in a separate folder for better file management.



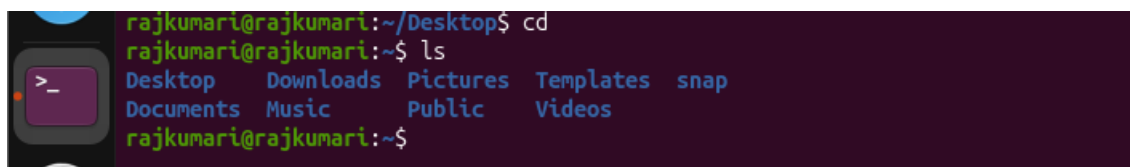
A terminal window titled 'rajkumari@rajkumari: ~/Desktop' with a dark purple background. The window shows the following commands and output:

```
rajkumari@rajkumari:~$ cd
rajkumari@rajkumari:~$ cd ~
rajkumari@rajkumari:~$ pwd
/home/rajkumari
rajkumari@rajkumari:~$ cd Desktop/
rajkumari@rajkumari:~/Desktop$ mkdir Tech_Hori
rajkumari@rajkumari:~/Desktop$ ls
Tech_Hori
rajkumari@rajkumari:~/Desktop$
```

4. How to List the contents of a directory.

Objective: Display the files and directories in the current or specified directory.

Purpose: View the contents of a directory to check available files and folders.



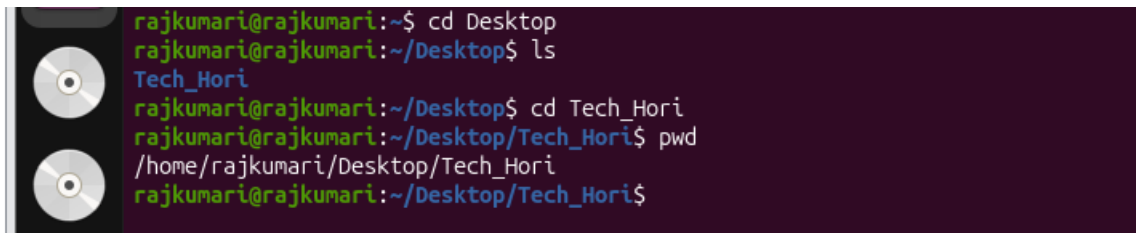
A terminal window showing the output of the 'ls' command. The window title is 'rajkumari@rajkumari: ~/Desktop'. The output lists the following directories:

```
rajkumari@rajkumari:~/Desktop$ cd
rajkumari@rajkumari:~$ ls
Desktop  Downloads  Pictures  Templates  snap
Documents Music      Public    Videos
rajkumari@rajkumari:~$
```

5. How to Change the current directory.

Objective: Change the current working directory to another directory.

Purpose: Navigate between directories to access and manage files and folders.

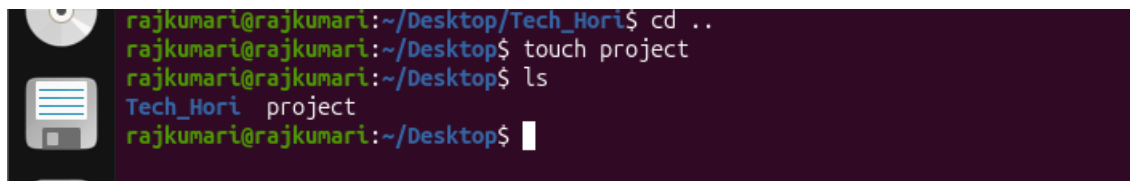
A terminal window with a dark purple background. On the left, there are two circular icons: the top one is a CD/DVD and the bottom one is a floppy disk. The terminal text shows a user named 'rajkumari' navigating from the home directory to the Desktop, then to a directory named 'Tech_Hori', and finally using 'pwd' to confirm the current path.

```
rajkumari@rajkumari:~$ cd Desktop
rajkumari@rajkumari:~/Desktop$ ls
Tech_Hori
rajkumari@rajkumari:~/Desktop$ cd Tech_Hori
rajkumari@rajkumari:~/Desktop/Tech_Hori$ pwd
/home/rajkumari/Desktop/Tech_Hori
rajkumari@rajkumari:~/Desktop/Tech_Hori$
```

6. How to Create a new empty file.

Objective: Create a new empty file.

Purpose: Create a file that can later be used to store text, data, or program code.

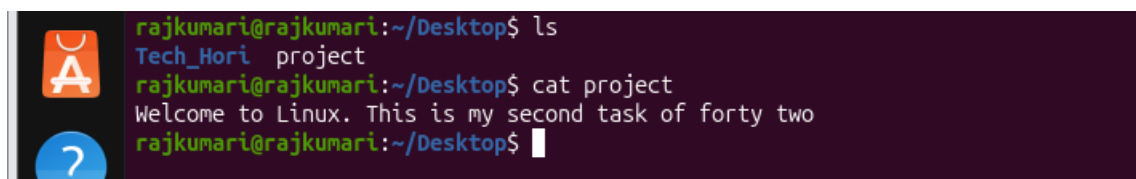
A terminal window with a dark purple background. On the left, there are two circular icons: the top one is a CD/DVD and the bottom one is a floppy disk. The terminal text shows the user navigating to the Desktop, creating a new file named 'project' using the 'touch' command, and then listing the files to confirm its creation.

```
rajkumari@rajkumari:~/Desktop/Tech_Hori$ cd ..
rajkumari@rajkumari:~/Desktop$ touch project
rajkumari@rajkumari:~/Desktop$ ls
Tech_Hori  project
rajkumari@rajkumari:~/Desktop$
```

7. How to View the contents of a file.

Objective: View the contents of a file.

Purpose: Display the text stored in a file without modifying it.

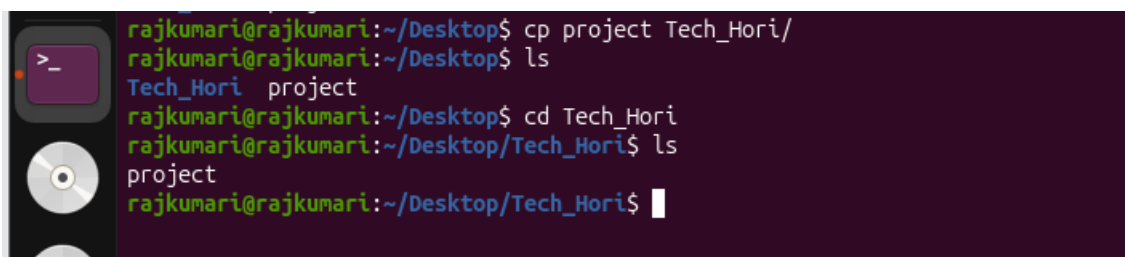
A terminal window with a dark purple background. On the left, there are two circular icons: the top one is a CD/DVD and the bottom one is a floppy disk. The terminal text shows the user listing files on the Desktop, then using the 'cat' command to view the contents of the 'project' file, which contains a welcome message.

```
rajkumari@rajkumari:~/Desktop$ ls
Tech_Hori  project
rajkumari@rajkumari:~/Desktop$ cat project
Welcome to Linux. This is my second task of forty two
rajkumari@rajkumari:~/Desktop$
```

8. How to Copy a file to another location.

Objective: Copy a file from one location to another.

Purpose: Create a duplicate of a file for backup, sharing, or use in a different directory.

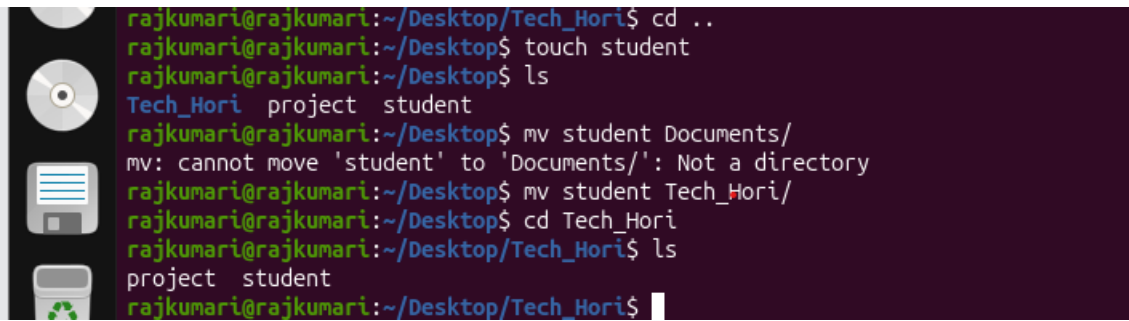
A terminal window with a dark purple background. On the left, there are two circular icons: the top one is a CD/DVD and the bottom one is a floppy disk. The terminal text shows the user copying the 'project' file to the 'Tech_Hori' directory using the 'cp' command, then navigating to that directory and listing the files to confirm the copy.

```
rajkumari@rajkumari:~/Desktop$ cp project Tech_Hori/
rajkumari@rajkumari:~/Desktop$ ls
Tech_Hori  project
rajkumari@rajkumari:~/Desktop$ cd Tech_Hori
rajkumari@rajkumari:~/Desktop/Tech_Hori$ ls
project
rajkumari@rajkumari:~/Desktop/Tech_Hori$
```

9. How to Move a file to another location.

Objective: Move a file from one location to another.

Purpose: Relocate a file to a different directory or rename it without creating a duplicate.

A terminal window with a dark purple background and light green text. On the left side, there are four circular icons: a CD, a floppy disk, a document, and a trash can. The terminal text shows a user named 'rajkumari' at a prompt 'rajkumari@rajkumari' in the directory '~/Desktop/Tech_Hori'. The user runs 'cd ..' to move to the desktop, then 'touch student' to create a file. A subsequent 'ls' command shows 'Tech_Hori', 'project', and 'student'. The user then attempts 'mv student Documents/' which fails with the message 'mv: cannot move 'student' to 'Documents/': Not a directory'. They then run 'mv student Tech_Hori/' successfully. Finally, they run 'cd Tech_Hori' and 'ls', which shows 'project' and 'student'.

10. How to Rename a file.

Objective: Rename an existing file.

Purpose: Change the name of a file without changing its contents or location.

```
rajkumari@rajkumari:~/Desktop/Tech_Hori$ ls
project student
rajkumari@rajkumari:~/Desktop/Tech_Hori$ mv student college
rajkumari@rajkumari:~/Desktop/Tech_Hori$ ls
college project
rajkumari@rajkumari:~/Desktop/Tech_Hori$
```

11. How to Delete a file.

Objective: Delete an existing file from the file system.

Purpose: Remove unwanted or unnecessary files to free up space and keep directories organized.

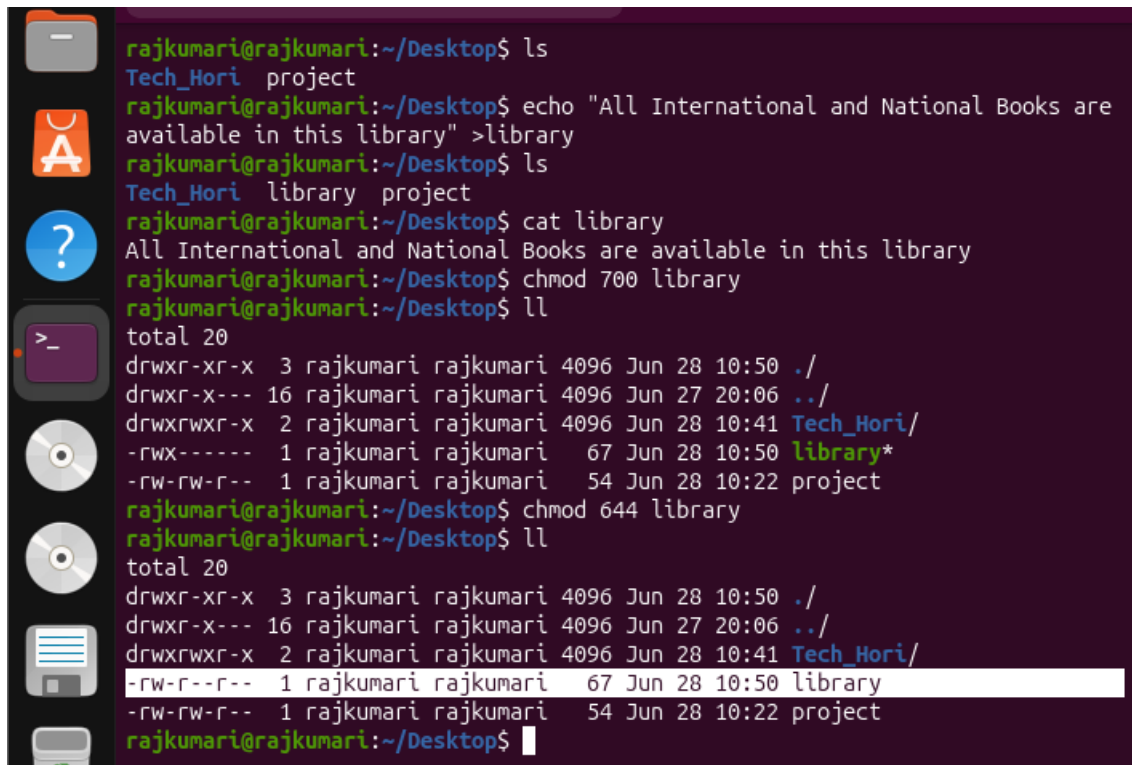
```
rajkumari@rajkumari:~/Desktop/Tech_Hori$ ls
college project
rajkumari@rajkumari:~/Desktop/Tech_Hori$ rm college
rajkumari@rajkumari:~/Desktop/Tech_Hori$ ls
project
rajkumari@rajkumari:~/Desktop/Tech_Hori$
```

12. How to grant or revoke permissions on a file or directory.

To **grant or revoke permissions** on a file or directory in Linux, use the **chmod** command.

Objective: Grant or revoke read, write, and execute permissions on a file or directory.

Purpose: Control who can access, modify, or execute files and directories, improving system security and access management.

A terminal window with a dark purple background. On the left, there is a vertical sidebar with icons for a file manager, application store, help, terminal, and other system utilities. The terminal text shows a user named 'rajkumari' at a machine named 'rajkumari' in the directory '~/Desktop'. The user runs 'ls' and sees 'Tech_Hori' and 'project'. Then they run 'echo "All International and National Books are available in this library" >library'. After another 'ls', they see 'Tech_Hori', 'library', and 'project'. They run 'cat library' and see the echo content. Then they run 'chmod 700 library'. Finally, they run 'll' (alias for 'ls -la') and see a detailed listing of files with permissions, owners, sizes, dates, and names. The 'library' file is highlighted in the output.

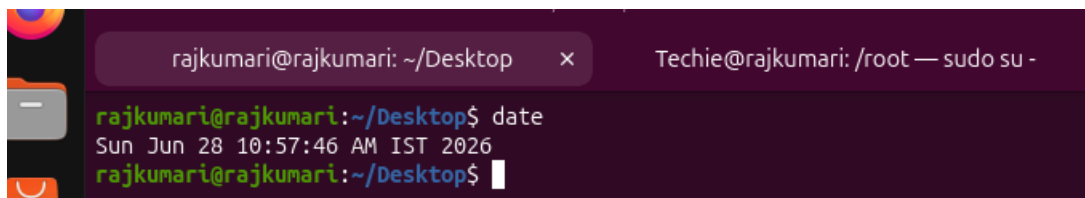
```
rajkumari@rajkumari:~/Desktop$ ls
Tech_Hori  project
rajkumari@rajkumari:~/Desktop$ echo "All International and National Books are
available in this library" >library
rajkumari@rajkumari:~/Desktop$ ls
Tech_Hori  library  project
rajkumari@rajkumari:~/Desktop$ cat library
All International and National Books are available in this library
rajkumari@rajkumari:~/Desktop$ chmod 700 library
rajkumari@rajkumari:~/Desktop$ ll
total 20
drwxr-xr-x  3 rajkumari rajkumari 4096 Jun 28 10:50 ./
drwxr-x--- 16 rajkumari rajkumari 4096 Jun 27 20:06 ../
drwxrwxr-x  2 rajkumari rajkumari 4096 Jun 28 10:41 Tech_Hori/
-rwx----- 1 rajkumari rajkumari   67 Jun 28 10:50 library*
-rw-rw-r--  1 rajkumari rajkumari   54 Jun 28 10:22 project
rajkumari@rajkumari:~/Desktop$ chmod 644 library
rajkumari@rajkumari:~/Desktop$ ll
total 20
drwxr-xr-x  3 rajkumari rajkumari 4096 Jun 28 10:50 ./
drwxr-x--- 16 rajkumari rajkumari 4096 Jun 27 20:06 ../
drwxrwxr-x  2 rajkumari rajkumari 4096 Jun 28 10:41 Tech_Hori/
-rw-r--r--  1 rajkumari rajkumari   67 Jun 28 10:50 library
-rw-rw-r--  1 rajkumari rajkumari   54 Jun 28 10:22 project
rajkumari@rajkumari:~/Desktop$
```

13. How to view the current date and time.

To **view the current date and time** in Linux, use the **date** command.

Objective: View the current system date and time.

Purpose: Check the system's current date, time, time zone, and day of the week.

A terminal window showing the 'date' command being executed. The terminal title bar shows two tabs: 'rajkumari@rajkumari: ~/Desktop' and 'Techie@rajkumari: /root — sudo su -'. The terminal text shows the user running 'date' and receiving the output 'Sun Jun 28 10:57:46 AM IST 2026'.

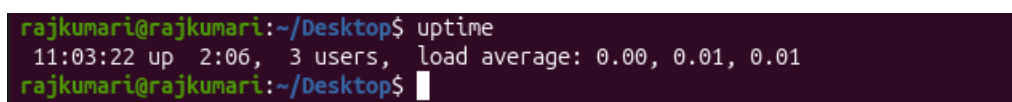
```
rajkumari@rajkumari:~/Desktop$ date
Sun Jun 28 10:57:46 AM IST 2026
rajkumari@rajkumari:~/Desktop$
```

14. How to check the system uptime.

To check the system uptime in Linux, use the **uptime** command.

Objective: Check how long the system has been running since the last boot.

Purpose: Monitor system uptime, current time, logged-in users, and system load averages.

A terminal window showing the 'uptime' command being executed. The terminal text shows the user running 'uptime' and receiving the output '11:03:22 up 2:06, 3 users, load average: 0.00, 0.01, 0.01'.

```
rajkumari@rajkumari:~/Desktop$ uptime
11:03:22 up 2:06, 3 users, load average: 0.00, 0.01, 0.01
rajkumari@rajkumari:~/Desktop$
```

15. How to view the running processes.

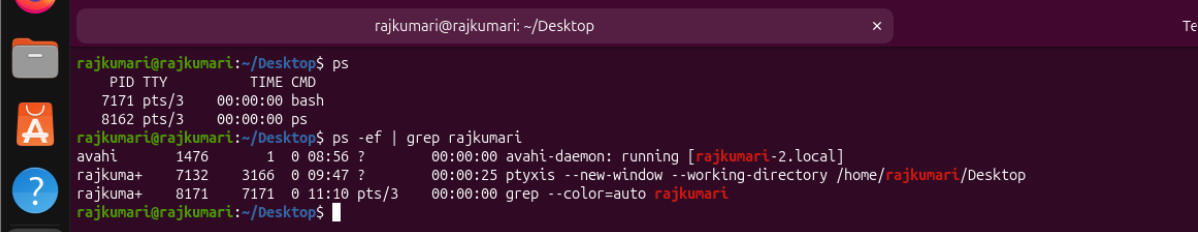
To view the running processes in Linux, use the **ps** command.

To view All the running process in Linux, use the **ps -ef** command.

To view specific running process, use the **ps -ef | grep username(rajkumari)** command.

Objective: View the processes that are currently running on the system.

Purpose: Monitor active processes, identify process IDs (PIDs), and troubleshoot system performance or application issues.



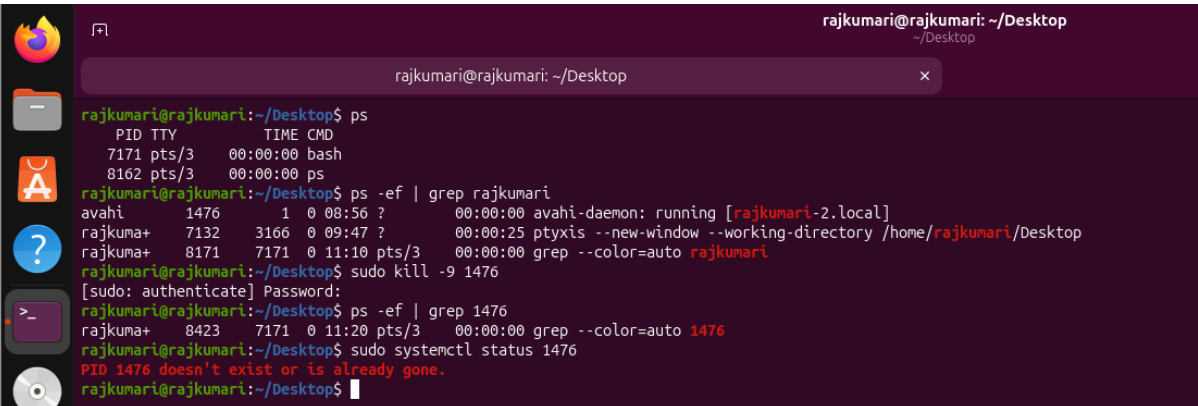
```
rajkumari@rajkumari: ~/Desktop
rajkumari@rajkumari:~/Desktop$ ps
  PID TTY          TIME CMD
  7171 pts/3    00:00:00 bash
  8162 pts/3    00:00:00 ps
rajkumari@rajkumari:~/Desktop$ ps -ef | grep rajkumari
avahi      1476      1  0 08:56 ?        00:00:00 avahi-daemon: running [rajkumari-2.local]
rajkuma+   7132    3166  0 09:47 ?        00:00:25 pttyx --new-window --working-directory /home/rajkumari/Desktop
rajkuma+   8171    7171  0 11:10 pts/3    00:00:00 grep --color=auto rajkumari
rajkumari@rajkumari:~/Desktop$
```

16. How to Kill a running process.

To kill a running process in Linux, use the **kill** command.

Objective: Terminate a running process using its Process ID (PID).

Purpose: Stop unresponsive, unnecessary, or resource-intensive processes to maintain system performance and stability.



```
rajkumari@rajkumari: ~/Desktop
rajkumari@rajkumari:~/Desktop$ ps
  PID TTY          TIME CMD
  7171 pts/3    00:00:00 bash
  8162 pts/3    00:00:00 ps
rajkumari@rajkumari:~/Desktop$ ps -ef | grep rajkumari
avahi      1476      1  0 08:56 ?        00:00:00 avahi-daemon: running [rajkumari-2.local]
rajkuma+   7132    3166  0 09:47 ?        00:00:25 pttyx --new-window --working-directory /home/rajkumari/Desktop
rajkuma+   8171    7171  0 11:10 pts/3    00:00:00 grep --color=auto rajkumari
rajkumari@rajkumari:~/Desktop$ sudo kill -9 1476
[sudo: authenticate] Password:
rajkumari@rajkumari:~/Desktop$ ps -ef | grep 1476
rajkuma+   8423    7171  0 11:20 pts/3    00:00:00 grep --color=auto 1476
rajkumari@rajkumari:~/Desktop$ sudo systemctl status 1476
PID 1476 doesn't exist or is already gone.
rajkumari@rajkumari:~/Desktop$
```

17. How to Install a package using the package manager (e.g., apt or yum).

To Install a package using the package manager on Ubuntu, use the **apt** package manager.

Objective: Install a software package using the Linux package manager.

Purpose: Download and install software along with its required dependencies, making it available for use on the system.

```
rajkumari@rajkumari: ~/Desktop
rajkumari@rajkumari:~/Desktop$ sudo apt update
Hit:1 http://in.archive.ubuntu.com/ubuntu resolute InRelease
Hit:2 http://in.archive.ubuntu.com/ubuntu resolute-updates InRelease
Hit:3 http://in.archive.ubuntu.com/ubuntu resolute-backports InRelease
Hit:4 http://security.ubuntu.com/ubuntu resolute-security InRelease
36 packages can be upgraded. Run 'apt list --upgradable' to see them.
rajkumari@rajkumari:~/Desktop$ sudo apt install git
Installing:
git
Installing dependencies:
git-man liberror-perl
Suggested packages:
git-doc git-email git-gui gitk gitweb git-cvs git-svn
Summary:
Upgrading: 0, Installing: 3, Removing: 0, Not Upgrading: 36
Download size: 5,453 kB
Space needed: 28.4 MB / 9,128 MB available
Continue? [Y/n] Y
Get:1 http://in.archive.ubuntu.com/ubuntu resolute/main amd64 liberror-perl all 0.17030-1 [23.5 kB]
Get:2 http://in.archive.ubuntu.com/ubuntu resolute/main amd64 git-man all 1:2.53.0-1ubuntu1 [1,202 kB]
Get:3 http://in.archive.ubuntu.com/ubuntu resolute/main amd64 git amd64 1:2.53.0-1ubuntu1 [4,227 kB]
Fetched 5,453 kB in 2min 21s (38.7 kB/s)
Selecting previously unselected package liberror-perl.
(Reading database... 155264 files and directories currently installed.)
Preparing to unpack .../liberror-perl_0.17030-1_all.deb...
Unpacking liberror-perl (0.17030-1)...
Selecting previously unselected package git-man.
Preparing to unpack .../git-man_1%3a2.53.0-1ubuntu1_all.deb...
Unpacking git-man (1:2.53.0-1ubuntu1)...
Selecting previously unselected package git.
Preparing to unpack .../git_1%3a2.53.0-1ubuntu1_amd64.deb...
Unpacking git (1:2.53.0-1ubuntu1)...
Setting up liberror-perl (0.17030-1)...
Setting up git-man (1:2.53.0-1ubuntu1)...
Setting up git (1:2.53.0-1ubuntu1)...
Processing triggers for man-db (2.13.1-1build1)...
rajkumari@rajkumari:~/Desktop$

rajkumari@rajkumari:~/Desktop$ git --version
git version 2.53.0
rajkumari@rajkumari:~/Desktop$
```

18. How to Update the system packages.

To update the system packages on Ubuntu, use the **apt/yum** package manager.

Objective: Update the system's package list and upgrade installed software packages.

Purpose: Keep the operating system secure, fix bugs, and install the latest software updates and security patches.

```
rajkumari@rajkumari:~/Desktop$ sudo apt update
Hit:1 http://security.ubuntu.com/ubuntu resolute-security InRelease
Hit:2 http://in.archive.ubuntu.com/ubuntu resolute InRelease
Hit:3 http://in.archive.ubuntu.com/ubuntu resolute-updates InRelease
Hit:4 http://in.archive.ubuntu.com/ubuntu resolute-backports InRelease
36 packages can be upgraded. Run 'apt list --upgradable' to see them.
rajkumari@rajkumari:~/Desktop$ sudo apt upgrade
Upgrading:
  accountsservice          libgtk-4-bin
  alsa-ucm-conf            libgtk-4-common
  base-files               libmalcontent-0-0
  cups-filters             libmalcontent-common
  cups-filters-core-drivers libmutter-18-0
  gir1.2-accountsservice-1.0 libpackagekit-glib2-18
  gir1.2-gtk-4.0           localsearch
  gir1.2-mutter-18         mutter-common
  gir1.2-packagekitglib-1.0 mutter-common-bin
  gnome-control-center     packagekit
  gnome-control-center-data python3-distupgrade
  gnome-control-center-faces resources
  gnome-session-bin       snapd
  gnome-session-common    ubuntu-release-upgrader-core
  gstreamer1.0-packagekit ubuntu-release-upgrader-gtk
  gtk-update-icon-cache   ubuntu-session
  libaccountsservice0     update-notifier
  libgtk-4-1              update-notifier-common

Summary:
Upgrading: 36, Installing: 0, Removing: 0, Not Upgrading: 0
Download size: 59.8 MB
Space needed: 651 kB / 9,102 MB available
```

19. How to Create a symbolic link.

In Linux, use the **ln -s** command to create a symbolic (soft) link.

Objective: Create a symbolic (soft) link to a file or directory.

Purpose: Provide an alternate path or shortcut to a file or directory without duplicating the original data.

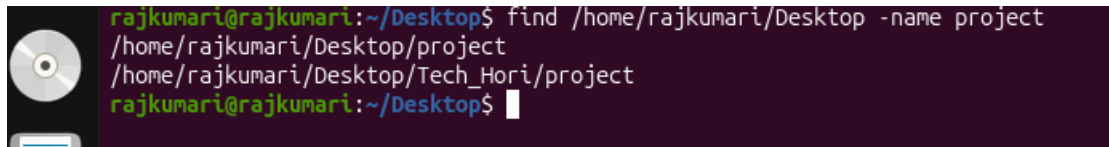
```
rajkumari@rajkumari: ~/Desktop | Techie@rajkumari: /root — s | rajkumari@rajkumari: ~/Desktop x
rajkumari@rajkumari:~/Desktop$ touch file.txt
rajkumari@rajkumari:~/Desktop$ ls
Tech_Hori file.txt library project
rajkumari@rajkumari:~/Desktop$ ln -s file1.txt file_link.txt
rajkumari@rajkumari:~/Desktop$ ls -l
total 12
drwxrwxr-x 2 rajkumari rajkumari 4096 Jun 28 10:41 Tech_Hori
-rw-rw-r-- 1 rajkumari rajkumari   0 Jun 28 12:46 file.txt
lrwxrwxrwx 1 rajkumari rajkumari   9 Jun 28 12:47 file_link.txt -> file1.txt
-rw-r--r-- 1 rajkumari rajkumari  67 Jun 28 10:50 library
-rw-rw-r-- 1 rajkumari rajkumari  54 Jun 28 10:22 project
rajkumari@rajkumari:~/Desktop$
```

20. How to Search for files using the find command.

Use the **find** command to search for files or directories based on name, type, size, permissions, and more.

Objective: Search for files or directories in the file system.

Purpose: Quickly locate files or directories by name, type, or other attributes, making file management easier.

A terminal window with a dark purple background. The prompt is 'rajkumari@rajkumari:~/Desktop\$'. The command 'find /home/rajkumari/Desktop -name project' is entered. The output shows two paths: '/home/rajkumari/Desktop/project' and '/home/rajkumari/Desktop/Tech_Hori/project'. The prompt is now 'rajkumari@rajkumari:~/Desktop\$' with a cursor.

```
rajkumari@rajkumari:~/Desktop$ find /home/rajkumari/Desktop -name project
/home/rajkumari/Desktop/project
/home/rajkumari/Desktop/Tech_Hori/project
rajkumari@rajkumari:~/Desktop$
```

21. How to Compress and decompress files using tar.

The **tar** command is used to **archive**, **compress**, and **extract** files and directories in Linux.

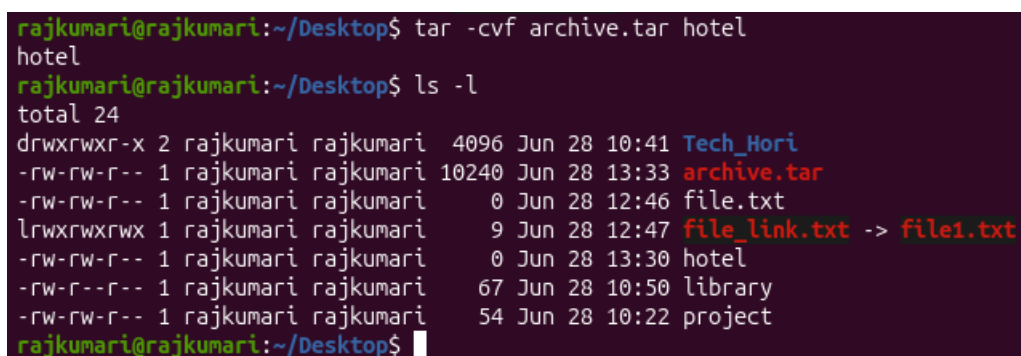
Objective: Compress files or directories into an archive and extract archived files using the tar command.

Purpose: Reduce storage space, combine multiple files into a single archive, simplify file transfer, and create backups.

Common Options

<u>Option</u>	<u>Meaning</u>
---------------	----------------

- | | |
|----|-----------------------------------|
| -c | Create a new archive |
| -x | Extract an archive |
| -v | Verbose (show progress) |
| -f | Specify the archive filename |
| -z | Compress or decompress using gzip |
| -t | List the contents of an archive |

A terminal window with a dark purple background. The prompt is 'rajkumari@rajkumari:~/Desktop\$'. The command 'tar -cvf archive.tar hotel' is entered, followed by 'hotel' on the next line. The prompt is now 'rajkumari@rajkumari:~/Desktop\$'. The command 'ls -l' is entered. The output shows a list of files and directories with their permissions, owner, group, size, date, and time. The files are: 'Tech_Hori' (directory), 'archive.tar' (file), 'file.txt' (file), 'file_link.txt -> file1.txt' (symlink), 'hotel' (directory), 'library' (file), and 'project' (file). The prompt is now 'rajkumari@rajkumari:~/Desktop\$' with a cursor.

```
rajkumari@rajkumari:~/Desktop$ tar -cvf archive.tar hotel
hotel
rajkumari@rajkumari:~/Desktop$ ls -l
total 24
drwxrwxr-x 2 rajkumari rajkumari 4096 Jun 28 10:41 Tech_Hori
-rw-rw-r-- 1 rajkumari rajkumari 10240 Jun 28 13:33 archive.tar
-rw-rw-r-- 1 rajkumari rajkumari    0 Jun 28 12:46 file.txt
lrwxrwxrwx 1 rajkumari rajkumari    9 Jun 28 12:47 file_link.txt -> file1.txt
-rw-rw-r-- 1 rajkumari rajkumari    0 Jun 28 13:30 hotel
-rw-r--r-- 1 rajkumari rajkumari  67 Jun 28 10:50 library
-rw-rw-r-- 1 rajkumari rajkumari  54 Jun 28 10:22 project
rajkumari@rajkumari:~/Desktop$
```

22. How to Monitor system resources with top or htop.

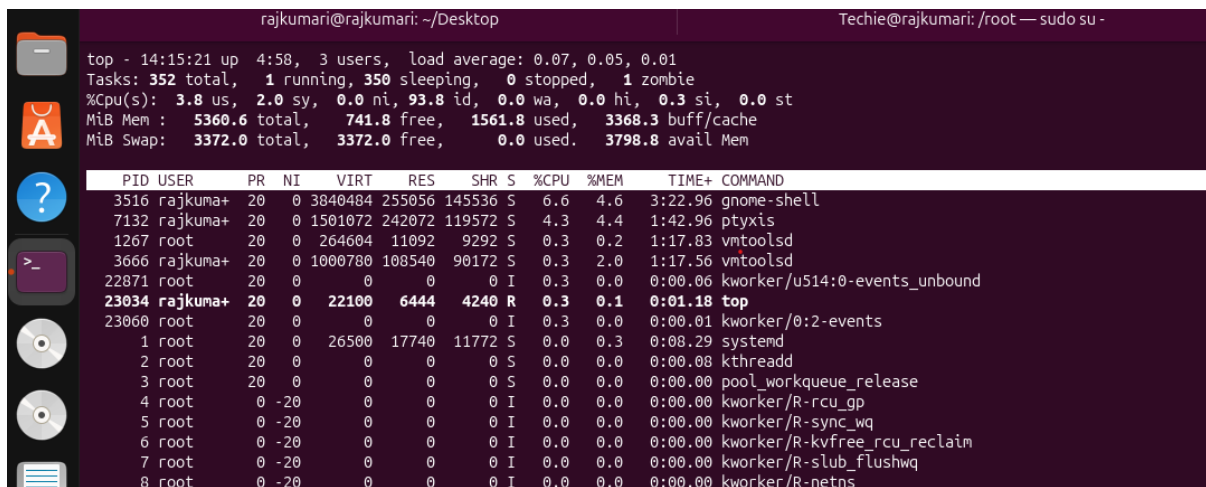
To monitor system resources in Linux, you can use either **top** or **htop**.

Objective: Monitor the system's CPU, memory, swap usage, and running processes in real time.

Purpose: To identify resource-intensive processes, troubleshoot performance issues, and monitor overall system health.

What it shows:

- CPU usage
- Memory (RAM) usage
- Swap memory usage
- Running processes
- Process IDs (PID)
- User owning each process
- CPU and memory usage per process



```
rajkumari@rajkumari: ~/Desktop
Techie@rajkumari: /root — sudo su -

top - 14:15:21 up 4:58, 3 users, load average: 0.07, 0.05, 0.01
Tasks: 352 total, 1 running, 350 sleeping, 0 stopped, 1 zombie
%Cpu(s): 3.8 us, 2.0 sy, 0.0 ni, 93.8 id, 0.0 wa, 0.0 hi, 0.3 si, 0.0 st
MiB Mem : 5360.6 total, 741.8 free, 1561.8 used, 3368.3 buff/cache
MiB Swap: 3372.0 total, 3372.0 free, 0.0 used, 3798.8 avail Mem

  PID USER      PR  NI    VIRT    RES    SHR  S  %CPU  %MEM   TIME+ COMMAND
 3516 rajkuma+  20   0 3840484 255056 145536 S   6.6   4.6   3:22.96 gnome-shell
 7132 rajkuma+  20   0 1501072 242072 119572 S   4.3   4.4   1:42.96 ptysis
 1267 root      20   0 264604 11092  9292 S   0.3   0.2   1:17.83 vntoolsd
 3666 rajkuma+  20   0 1000780 108540 90172 S   0.3   2.0   1:17.56 vntoolsd
22871 root      20   0      0      0      0 I   0.3   0.0   0:00.06 kworker/u514:0-events_unbound
23034 rajkuma+  20   0 22100 6444 4240 R   0.3  0.1  0:01.18 top
23060 root      20   0      0      0      0 I   0.3   0.0   0:00.01 kworker/0:2-events
    1 root      20   0 26500 17740 11772 S   0.0   0.3   0:08.29 systemd
    2 root      20   0      0      0      0 S   0.0   0.0   0:00.08 kthreadd
    3 root      20   0      0      0      0 S   0.0   0.0   0:00.00 pool_workqueue_release
    4 root      0 -20      0      0      0 I   0.0   0.0   0:00.00 kworker/R-rcu_gp
    5 root      0 -20      0      0      0 I   0.0   0.0   0:00.00 kworker/R-sync_wq
    6 root      0 -20      0      0      0 I   0.0   0.0   0:00.00 kworker/R-kvfree_rcu_reclaim
    7 root      0 -20      0      0      0 I   0.0   0.0   0:00.00 kworker/R-slub_flushwq
    8 root      0 -20      0      0      0 I   0.0   0.0   0:00.00 kworker/R-netns
```

23. How to create and manage user groups.

Objective: To create, modify, and manage user groups for controlling user permissions and access.

Purpose: Groups allow multiple users to share the same permissions on files and directories, making user administration easier.

Task	Command
Create a new group	\$ sudo groupadd filename

Task	Command
Verify the group exists	\$ getent group filename
Add a user to a group	\$ sudo usermod -aG filename username
View the groups a user belongs to	\$ groups username
View the current user's groups	\$ groups
Remove a user from a group	\$ sudo gpasswd -d username filename
Change a user's primary group	\$ sudo usermod -g filename username
Rename a group	\$ sudo groupmod -n devteam filename
Delete a group	\$ sudo groupdel devteam
List all groups	\$ cat /etc/group

```

rajkumari@rajkumari:/root$ sudo usermod -aG developers Techie
rajkumari@rajkumari:/root$ group Techie
Command 'group' not found, did you mean:
  command 'groups' from deb coreutils-from-gnu (0.0.0~ubuntu25)
  command 'groups' from deb coreutils-from-uutils (0.0.0~ubuntu25)
  command 'groups' from deb coreutils-from-busybox (0.0.0~ubuntu25)
  command 'groups' from deb coreutils-from-toybox (0.0.0~ubuntu25)
Try: sudo apt install <deb name>
rajkumari@rajkumari:/root$ sudo group Techie
sudo: 'group': command not found
rajkumari@rajkumari:/root$ sudo groups Techie
Techie : Techie sudo users developers
rajkumari@rajkumari:/root$

```

24. How to set up SSH password less authentication.

Objective: To configure SSH key-based authentication so that users can log in to a remote server without entering a password.

Purpose: SSH password-less authentication improves security and simplifies remote access by using public/private key pairs instead of passwords.

MINGW64:/c/Users/braj

```
brajk@RajKumari MINGW64 ~
$ cat ~/.ssh/id_rsa.pub
ssh-rsa AAAAB3NzaC1yc2EAAAADAQABAAQCAQCj5jsCdVZjKNVwgZLN6YeifGkwPp4NWq+4DwyWRmBj
3/gFoU91hNr+nuWshG1pj/wDGVntUMCGjPTRfQqnxsvhezbb6r9PQsvADtws6WHNlZPRYyCuSZVUizYrL
B14UXICOZyV9zn8PwXZ5iq5kDZX2ctkEhiZ3PgnSiCHk2gQ4u5G1S9WsKwO9A6T0kyr438Cx7r+/GRpn
By6c3dj/GAov1Jegng6GtTfc6pMkqy5jji0e7ZF89BYMvmky5sfoUG6Z5ErwV33TJHiOuvze3qCeT6Sg
fZaGwrhCxKijsj55WAckig8kbqna1Sig/grG9Yo4fQJGI+UNE019bQNN08xdCfKi2sLLsqMxnjmuRYA7
98JgV28xhMiqc5L2Hu8wwI2A1yb42L003QeQ3kXsIqL5JQxchqALJp8xQxcI5DiArM8C6VxpakL1pwhd
2uc8DmNowVoBUml0oasj9zYPxd3EDma6Lklz16/aqZVWN8151MjiAub9mu2lpo1U7xjHLhtpZADHBStJ
Y5pU/Xn3tAV1XA9HpR7cFAAHoFyiIW71tr5gpR7LhjQ2g9NCfPKCwt5hKoN/4dJ8Jy3cgU29XMwUna2X
UMUofDhY4Kyft4Mpd96DDDPYTYA4owzdRIK7M7BRK4/3ePXy+RBnSjL51b9+7yK02DnNTZUsium4Tf9X
pw== brajk@RajKumari

brajk@RajKumari MINGW64 ~
```

 ec2-user@ip-172-31-28-28:~\$

```
brajk@RajKumari MINGW64 ~/Downloads
$ ssh -i linux-testing.pem ec2-user@3.87.146.30

#_
##### Amazon Linux 2023
~\#####\
~\####|
~\##|
~\#/ https://aws.amazon.com/linux/amazon-linux-2023
V'-'->
~
~.-.-
~/m/'

Last login: Sun Jun 28 09:41:32 2026 from 103.248.210.235
[ec2-user@ip-172-31-28-28 ~]$ mkdir -p ~/.ssh
chmod 700 ~/.ssh
[ec2-user@ip-172-31-28-28 ~]$ vi ~/.ssh/authorized_keys
[ec2-user@ip-172-31-28-28 ~]$ chmod 600 ~/.ssh/authorized_keys
[ec2-user@ip-172-31-28-28 ~]$ ssh ec2-user@3.87.146.30
The authenticity of host '3.87.146.30 (3.87.146.30)' can't be established.
ED25519 key fingerprint is SHA256:EMBFLLrzpuDonknqxuBE08dxRsEaSlSFHKbq8w0fnQ1I.
This key is not known by any other names
Are you sure you want to continue connecting (yes/no/[fingerprint])? n
Please type 'yes', 'no' or the fingerprint: no
Host key verification failed.
[ec2-user@ip-172-31-28-28 ~]$ vi ~/.ssh/authorized_keys
[ec2-user@ip-172-31-28-28 ~]$ chmod 600 ~/.ssh/authorized_keys
[ec2-user@ip-172-31-28-28 ~]$ exit
logout
Connection to 3.87.146.30 closed.
```

[illegible]

25. How to monitor log files using tail or grep.

Objective: To monitor and analyze Linux log files in real time using the tail and grep commands for system administration and troubleshooting.

Purpose: To view the latest log entries and monitor system activities as they occur.

- To search log files for specific events, errors, or keywords for debugging and security analysis.
- To identify and troubleshoot system, application, or SSH-related issues quickly.

```
rajkumari@rajkumari:/root$ cd
rajkumari@rajkumari:~$ cd Desktop
rajkumari@rajkumari:~/Desktop$ tail -f /var/log/syslog
2026-06-28T15:43:09.157805+05:30 rajkumari gnome-shell[3516]: Gio.DBusError: GDBus.Error:org.freedesktop.DBus.Error.Failed: Set global engine failed: Open
2026-06-28T15:43:09.415858+05:30 rajkumari gnome-shell[3516]: DING: Detected async api for thumbnails
2026-06-28T15:43:09.512209+05:30 rajkumari gnome-shell[3516]: DING: ** Message: 15:43:09.511: Connecting to org.freedesktop.Tracker3.Miner.Files
2026-06-28T15:43:09.517912+05:30 rajkumari gnome-shell[3516]: DING: GNOME nautilus 50.0
2026-06-28T15:43:09.549757+05:30 rajkumari gnome-shell[3516]: DING: DBus interface for Switcheroo control (net.hadess.SwitcherooControl) is now available.
2026-06-28T15:43:09.550913+05:30 rajkumari gnome-shell[3516]: DING: DBus interface for Nautilus (org.gnome.Nautilus.FileOperations2) is now available.
2026-06-28T15:43:09.551002+05:30 rajkumari gnome-shell[3516]: DING: DBus interface for Gvfs daemon (org.gtk.vfs.Metadata) is now available.
2026-06-28T15:43:09.551033+05:30 rajkumari gnome-shell[3516]: DING: DBus interface for Nautilus (org.freedesktop.FileManager1) is now available.
2026-06-28T15:43:09.577775+05:30 rajkumari gnome-shell[3516]: Received notification for window. 0 notifications remaining.
2026-06-28T15:43:36.460889+05:30 rajkumari systemd[1]: fprintd.service: Deactivated successfully.
rajkumari@rajkumari:~/Desktop$ tail -f /var/log/syslog
2026-06-28T15:43:09.551002+05:30 rajkumari gnome-shell[3516]: DING: DBus interface for Gvfs daemon (org.gtk.vfs.Metadata) is now available.
2026-06-28T15:43:09.551033+05:30 rajkumari gnome-shell[3516]: DING: DBus interface for Nautilus (org.freedesktop.FileManager1) is now available.
2026-06-28T15:43:09.577775+05:30 rajkumari gnome-shell[3516]: Received notification for window. 0 notifications remaining.
2026-06-28T15:43:36.460889+05:30 rajkumari systemd[1]: fprintd.service: Deactivated successfully.
2026-06-28T15:45:27.854843+05:30 rajkumari systemd[3166]: Started pttyx-spawn-5755d0d4-d392-4cc1-a6f7-68b37f940ad1.scope - [systemd-run] /usr/bin/bash.
rajkumari@rajkumari:~/Desktop$ grep ssh /var/log/syslog
2026-06-27T12:57:38.879188+00:00 rajkumari systemd[1]: sssd-ssh.socket: Bound to unit sssd.service, but unit isn't active.
2026-06-27T12:57:38.879191+00:00 rajkumari systemd[1]: Dependency failed for sssd-ssh.socket - SSSD SSH Service responder socket.
2026-06-27T12:57:38.879206+00:00 rajkumari systemd[1]: sssd-ssh.socket: Job sssd-ssh.socket/start failed with result 'dependency'.
2026-06-27T12:57:43.285790+00:00 rajkumari systemd[2034]: Starting ssh-agent.socket - OpenSSH Agent socket...
2026-06-27T12:57:43.308774+00:00 rajkumari systemd[2034]: Listening on ssh-agent.socket - OpenSSH Agent socket.
2026-06-27T12:57:43.311178+00:00 rajkumari systemd[2034]: Starting gcr-ssh-agent.socket - GCR ssh-agent wrapper...
2026-06-27T12:57:43.329382+00:00 rajkumari systemd[2034]: Listening on gcr-ssh-agent.socket - GCR ssh-agent wrapper.
2026-06-27T12:57:43.331492+00:00 rajkumari systemd[2034]: Starting gpg-agent-ssh.socket - GnuPG cryptographic agent (ssh-agent emulation)...
2026-06-27T12:57:43.354275+00:00 rajkumari systemd[2034]: Listening on gpg-agent-ssh.socket - GnuPG cryptographic agent (ssh-agent emulation).
2026-06-27T12:57:43.568460+00:00 rajkumari systemd[2034]: Started gcr-ssh-agent.service - GCR ssh-agent wrapper.
2026-06-27T12:57:43.585360+00:00 rajkumari systemd[2034]: Started ssh-agent.service - OpenSSH Agent.
2026-06-27T12:57:58.858642+00:00 rajkumari cloud-init[866]: Your identification has been saved in /etc/ssh/ssh_host_rsa_key
2026-06-27T12:57:58.858663+00:00 rajkumari cloud-init[866]: Your public key has been saved in /etc/ssh/ssh_host_rsa_key.pub
2026-06-27T12:57:58.859015+00:00 rajkumari cloud-init[866]: Your identification has been saved in /etc/ssh/ssh_host_ecdsa_key
2026-06-27T12:57:58.859036+00:00 rajkumari cloud-init[866]: Your public key has been saved in /etc/ssh/ssh_host_ecdsa_key.pub
2026-06-27T12:57:58.859394+00:00 rajkumari cloud-init[866]: Your identification has been saved in /etc/ssh/ssh_host_ed25519_key
2026-06-27T12:57:58.859415+00:00 rajkumari cloud-init[866]: Your public key has been saved in /etc/ssh/ssh_host_ed25519_key.pub
2026-06-27T12:57:59.345123+00:00 rajkumari systemd[3181]: Starting ssh-agent.socket - OpenSSH Agent socket...
```

26. how to set up a web server (e.g., Apache or Nginx).

Objective: To install, configure, and run the Apache web server on Ubuntu, enabling the system to host and serve web pages over HTTP.

Purpose: To host websites or web applications on a Linux server.

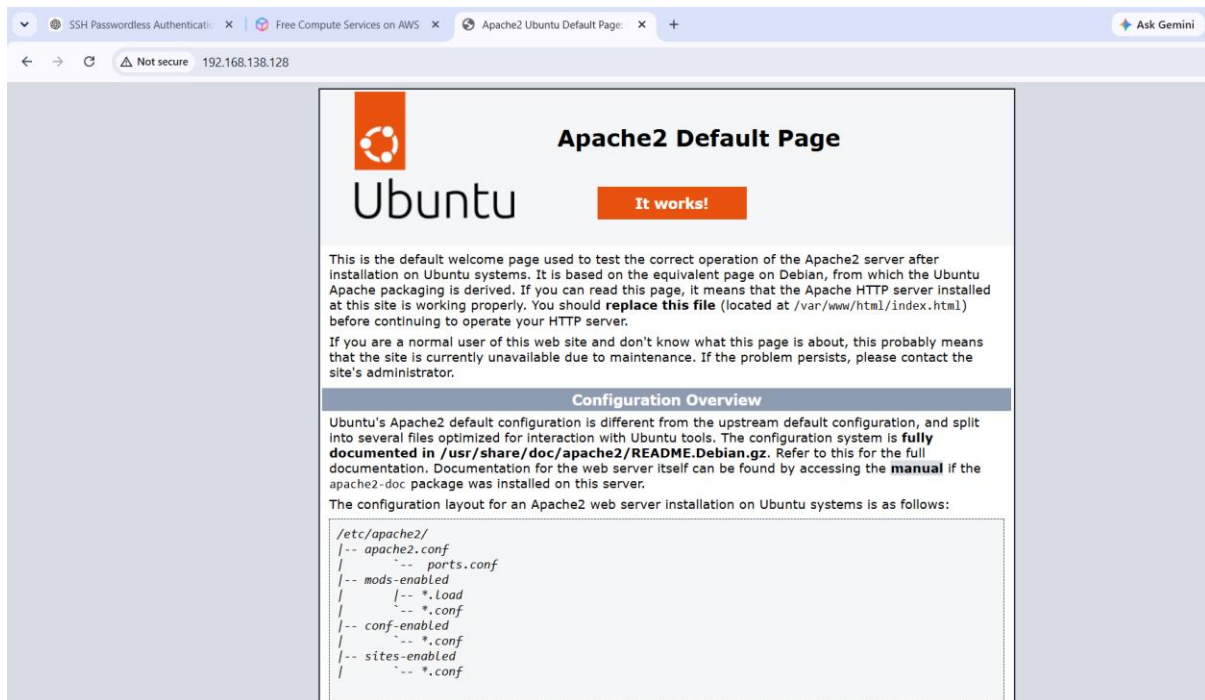
- To learn basic web server installation and configuration.
- To verify web server accessibility through a web browser.

```
rajkumari@rajkumari: ~/Desktop | Techie@rajkumari: /root — sudo su - | rajkumari@rajkumari: ~ — sudo : x | rajkumari@rajkumari: ~/Desktop | rajkumari@rajkumari: ~/Desktop$ cd
rajkumari@rajkumari:~$ sudo apt update
[sudo: authenticate] Password:
Hit:1 http://security.ubuntu.com/ubuntu resolute-security InRelease
Hit:2 http://in.archive.ubuntu.com/ubuntu resolute InRelease
Hit:3 http://in.archive.ubuntu.com/ubuntu resolute-updates InRelease
Hit:4 http://in.archive.ubuntu.com/ubuntu resolute-backports InRelease
All packages are up to date.
rajkumari@rajkumari:~$ sudo apt install apache2 -y
Installing:
  apache2
Installing dependencies:
  apache2-bin      apache2-utils  libaprutil1-dbd-sqlite3  libaprutil1t64
  apache2-data     libapr1t64     libaprutil1-ldap
Suggested packages:
  apache2-doc  apache2-suexec-pristine | apache2-suexec-custom
Summary:
  Upgrading: 0, Installing: 8, Removing: 0, Not Upgrading: 0
  Download size: 1,935 kB
  Space needed: 7,565 kB / 8,814 MB available
Get:1 http://in.archive.ubuntu.com/ubuntu resolute/main amd64 libapr1t64 amd64 1.7.6-3 [109 kB]
Get:2 http://in.archive.ubuntu.com/ubuntu resolute/main amd64 libaprutil1t64 amd64 1.6.3-3ubuntu3 [93.8 kB]
Get:3 http://in.archive.ubuntu.com/ubuntu resolute/main amd64 libaprutil1-dbd-sqlite3 amd64 1.6.3-3ubuntu3 [11.4 kB]
Get:4 http://in.archive.ubuntu.com/ubuntu resolute/main amd64 libaprutil1-ldap amd64 1.6.3-3ubuntu3 [9,226 B]
Get:5 http://in.archive.ubuntu.com/ubuntu resolute-updates/main amd64 apache2-bin amd64 2.4.66-2ubuntu2.2 [1,354 kB]
Get:6 http://in.archive.ubuntu.com/ubuntu resolute-updates/main amd64 apache2-data all 2.4.66-2ubuntu2.2 [163 kB]
Get:7 http://in.archive.ubuntu.com/ubuntu resolute-updates/main amd64 apache2-utils amd64 2.4.66-2ubuntu2.2 [93.8 kB]
Get:8 http://in.archive.ubuntu.com/ubuntu resolute-updates/main amd64 apache2 amd64 2.4.66-2ubuntu2.2 [100 kB]
Fetched 1,935 kB in 17s (117 kB/s)
Selecting previously unselected package libapr1t64:amd64.
(Reading database... 156416 files and directories currently installed.)
Preparing to unpack .../0-libapr1t64_1.7.6-3_amd64.deb...
Unpacking libapr1t64:amd64 (1.7.6-3)...
Selecting previously unselected package libaprutil1t64:amd64.
Preparing to unpack .../1-libaprutil1t64_1.6.3-3ubuntu3_amd64.deb...
Unpacking libaprutil1t64:amd64 (1.6.3-3ubuntu3)...
Selecting previously unselected package libaprutil1-dbd-sqlite3:amd64.
Preparing to unpack .../2-libaprutil1-dbd-sqlite3_1.6.3-3ubuntu3_amd64.deb...
```

```
rajkumari@rajkumari: ~/Desktop | Techie@rajkumari: /root — sudo su - | rajkumari@rajkumari: ~ — sudo : x | rajkumari@rajkumari: ~/Desktop | rajkumari@rajkumari: ~/Desktop$ cd
rajkumari@rajkumari:~$ sudo systemctl status apache2
[sudo: authenticate] Password:
● apache2.service - The Apache HTTP Server
   Loaded: loaded (/usr/lib/systemd/system/apache2.service; enabled; preset=
   Active: active (running) since Sun 2026-06-28 16:04:43 IST; 35s ago
   Invocation: d0844794ba2f48ef91158af7df7456c9
     Docs: https://httpd.apache.org/docs/2.4/
    Main PID: 25630 (apache2)
     Status: "Total requests: 0; Idle/Busy workers 100/0;Requests/sec: 0; Byt
      Tasks: 55 (limit: 3814)
     Memory: 5.6M (peak: 5.8M)
        CPU: 84ms
     CGroup: /system.slice/apache2.service
            └─25630 /usr/sbin/apache2 -k start -DFOREGROUND
              └─25646 /usr/sbin/apache2 -k start -DFOREGROUND
                └─25647 /usr/sbin/apache2 -k start -DFOREGROUND

Jun 28 16:04:43 rajkumari systemd[1]: Starting apache2.service - The Apache H
Jun 28 16:04:43 rajkumari apachectl[25630]: AH00558: apache2: Could not relia
Jun 28 16:04:43 rajkumari systemd[1]: Started apache2.service - The Apache HT
lines 1-18/18 (END)
```

```
rajkumari@rajkumari: ~/Desktop | Techie@rajkumari: /root — sudo su - | rajkumari@rajkumari: ~ — sudo : x | rajkumari@rajkumari: ~/Desktop | rajkumari@rajkumari: ~/Desktop$ hostname -I
192.168.138.128
```



27. How to configure and secure a MySQL Database.

Objective: To install, configure, and secure the MySQL database server on Ubuntu to protect it from unauthorized access and ensure safe database operations.

Purpose: Install and configure a MySQL database server.

- Secure the MySQL installation by removing insecure default settings.
- Create a database and user with appropriate privileges.
- Verify that the MySQL server is running properly.

```
rajkumari@rajkumari:~$ sudo systemctl status mysql
● mysql.service - MySQL Community Server
   Loaded: loaded (/usr/lib/systemd/system/mysql.service; enabled; preset: 
   Active: active (running) since Sun 2026-06-28 16:26:28 IST; 1min 40s ago
   Invocation: 58eba7eb44bf4a7588b972d227459043
   Process: 28780 ExecStartPre=/usr/share/mysql/mysql-systemd-start pre (cod
   Main PID: 28790 (mysqld)
   Status: "Server is operational"
   Tasks: 34 (limit: 3814)
   Memory: 478.9M (peak: 479.2M)
   CPU: 3.005s
   CGroup: /system.slice/mysql.service
           └─28790 /usr/sbin/mysqld

Jun 28 16:26:27 rajkumari systemd[1]: Starting mysql.service - MySQL Communit
Jun 28 16:26:28 rajkumari systemd[1]: Started mysql.service - MySQL Community
rajkumari@rajkumari:~$
```

```
rajkumari@rajkumari:~$ mysql --version
mysql Ver 8.4.10-0ubuntu0.26.04.1 for Linux on x86_64 ((Ubuntu))
rajkumari@rajkumari:~$
```

```
rajkumari@rajkumari:~$ mysql --version
mysql Ver 8.4.10-0ubuntu0.26.04.1 for Linux on x86_64 ((Ubuntu))
rajkumari@rajkumari:~$ sudo systemctl start mysql
rajkumari@rajkumari:~$ sudo systemctl enable mysql
rajkumari@rajkumari:~$ sudo systemctl is-enabled mysql
enabled
rajkumari@rajkumari:~$ sudo mysql_secure_installation

Securing the MySQL server deployment.

Connecting to MySQL using a blank password.

VALIDATE PASSWORD COMPONENT can be used to test passwords
and improve security. It checks the strength of password
and allows the users to set only those passwords which are
secure enough. Would you like to setup VALIDATE PASSWORD component?

Press y|Y for Yes, any other key for No: y

There are three levels of password validation policy:

LOW      Length >= 8
MEDIUM  Length >= 8, numeric, mixed case, and special characters
STRONG Length >= 8, numeric, mixed case, special characters and dictionary
        file

Please enter 0 = LOW, 1 = MEDIUM and 2 = STRONG: 2
```

```
All done!
rajkumari@rajkumari:~$ sudo mysql
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 10
Server version: 8.4.10-0ubuntu0.26.04.1 (Ubuntu)

Copyright (c) 2000, 2026, Oracle and/or its affiliates.

Oracle is a registered trademark of Oracle Corporation and/or its
affiliates. Other names may be trademarks of their respective
owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql> CREATE DATABASE companydb;
Query OK, 1 row affected (0.01 sec)

mysql> show databases;
-> show databases;
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version for the right syntax to use near 'show databases' at line 2
mysql> show databases;
+-----+
| Database |
+-----+
| companydb |
| information_schema |
| mysql |
| performance_schema |
| sys |
+-----+
5 rows in set (0.01 sec)

mysql>
```

28. How to Set up a Application Server (e.g.,Apache Tomcat)

Objective: To install, configure, and run the Apache Tomcat application server on Ubuntu to deploy and host Java web applications.

Purpose: Install Java (JDK), which is required for Tomcat.

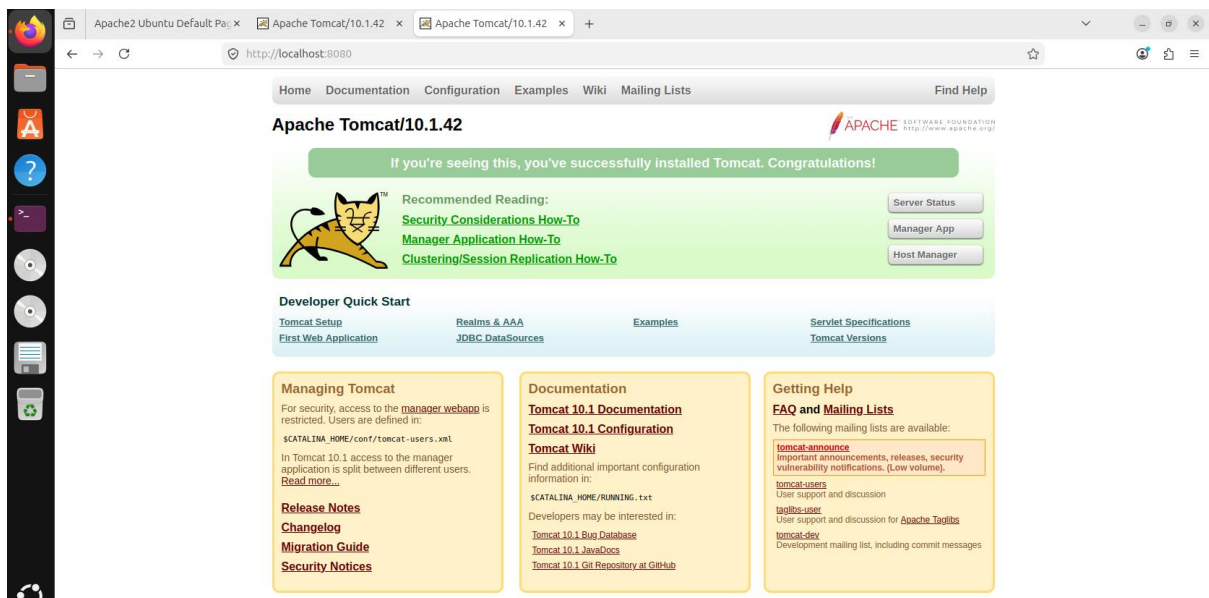
- Install and configure Apache Tomcat.
- Start and manage the Tomcat service.

- Access the Tomcat web interface through a browser.
- Deploy Java web applications (.war files).

```

Techie@rajkumari:/root — sudo su -
rajkumari@rajkumari: ~ — sudo su -
rajkumari@rajkumari: ~
rajkumari@rajkumari:~$ java -version
openjdk version "17.0.19" 2026-04-21
OpenJDK Runtime Environment (build 17.0.19+10-1-26.04.2-Ubuntu)
OpenJDK 64-Bit Server VM (build 17.0.19+10-1-26.04.2-Ubuntu, mixed mode, sharing)
rajkumari@rajkumari:~$ sudo /opt/tomcat/bin/startup.sh
[sudo: authenticate] Password:
Using CATALINA_BASE:   /opt/tomcat
Using CATALINA_HOME:   /opt/tomcat
Using CATALINA_TMPDIR: /opt/tomcat/temp
Using JRE_HOME:        /usr
Using CLASSPATH:        /opt/tomcat/bin/bootstrap.jar:/opt/tomcat/bin/tomcat-juli.jar
Tomcat started.
rajkumari@rajkumari:~$ ps -ef | grep tomcat
root      29591   3166  17 16:47 ?        00:00:02 /usr/bin/java -Djava.util.logging.config.file=/opt/tomcat/conf/logging.properties -Djava.util.logging.manager=org.apache.juli.ClassLoaderLogManager -Djdk.tls.ephemeralDHKeySize=2048 -Djava.protocol.handler.pkgs=org.apache.catalina.webresources -Dsun.io.useCanonCaches=false -Dorg.apache.catalina.security.SecurityListener.Umask=0027 --add-opens=java.base/java.lang=ALL-UNNAMED --add-opens=java.base/java.lang.reflect=ALL-UNNAMED --add-opens=java.base/java.io=ALL-UNNAMED --add-opens=java.base/java.util.concurrent=ALL-UNNAMED --add-opens=java.rmi/sun.rmi.transport=ALL-UNNAMED -classpath /opt/tomcat/bin/bootstrap.jar:/opt/tomcat/bin/tomcat-juli.jar -Dcatalina.base=/opt/tomcat -Dcatalina.home=/opt/tomcat -jar /opt/tomcat/bin/tomcat.jar
rajkumari@rajkumari:~$ sudo ss -tulpn | grep 8080
tcp        LISTEN 0      100      *:8080      *:8080      users:((("java",pid=29591,fd=43))
rajkumari@rajkumari:~$ sudo ufw allow 8080/tcp
Rules updated
Rules reloaded
Firewall not enabled (skipping reload)
rajkumari@rajkumari:~$ hostname -I
192.168.138.128
rajkumari@rajkumari:~$

```



29. How to create a service file for Apache Tomcat.(Should execute by using **systemctl** command)

To manage **Apache Tomcat using systemctl**, you need to create a **systemd service file**.

Objective: To create a systemd service for Apache Tomcat so it can be started, stopped, restarted, and enabled at boot using the systemctl command.

Purpose: Manage Tomcat as a Linux service.

- Automatically start Tomcat during system boot.
- Simplify Tomcat administration using systemctl.

```

rajkumari@rajkumari:~$ sudo nano /etc/systemd/system/tomcat.service
[sudo: authenticate] Password:
rajkumari@rajkumari:~$ sudo systemctl daemon-reload
sudo systemctl reset-failed
sudo systemctl restart tomcat
sudo systemctl status tomcat
● tomcat.service - Apache Tomcat
   Loaded: loaded (/etc/systemd/system/tomcat.service; enabled; preset: ena>
   Active: active (running) since Sun 2026-06-28 17:57:58 IST; 46ms ago
  Invocation: ba9fc93caf454048885726f88fc74c25
    Process: 32361 ExecStart=/opt/tomcat/bin/startup.sh (code=exited, status=>
   Main PID: 32371 (java)
      Tasks: 16 (limit: 3814)
     Memory: 16M (peak: 16M)
        CPU: 74ms
    CGroup: /system.slice/tomcat.service
           └─32371 /usr/lib/jvm/java-17-openjdk-amd64/bin/java -Djava.util.>

Jun 28 17:57:58 rajkumari systemd[1]: Starting tomcat.service - Apache Tomcat>
Jun 28 17:57:58 rajkumari startup.sh[32361]: Existing PID file found during s>
Jun 28 17:57:58 rajkumari startup.sh[32361]: Removing/clearing stale PID file.
Jun 28 17:57:58 rajkumari startup.sh[32361]: Tomcat started.
Jun 28 17:57:58 rajkumari systemd[1]: Started tomcat.service - Apache Tomcat.
rajkumari@rajkumari:~$ sudo systemctl start tomcat
sudo systemctl stop tomcat
sudo systemctl restart tomcat
sudo systemctl status tomcat
sudo systemctl enable tomcat
● tomcat.service - Apache Tomcat
   Loaded: loaded (/etc/systemd/system/tomcat.service; enabled; preset: ena>
   Active: active (running) since Sun 2026-06-28 17:58:35 IST; 22ms ago
  Invocation: cb1163618e75477e9c36b32bd58039f2
    Process: 32734 ExecStart=/opt/tomcat/bin/startup.sh (code=exited, status=>
   Main PID: 32744 (java)
      Tasks: 7 (limit: 3814)
     Memory: 13.4M (peak: 13.4M)
        CPU: 60ms
    CGroup: /system.slice/tomcat.service
           └─32744 /usr/lib/jvm/java-17-openjdk-amd64/bin/java -Djava.util.>

Jun 28 17:58:35 rajkumari systemd[1]: Starting tomcat.service - Apache Tomcat>
Jun 28 17:58:35 rajkumari startup.sh[32734]: Existing PID file found during s>
Jun 28 17:58:35 rajkumari startup.sh[32734]: Removing/clearing stale PID file.
Jun 28 17:58:35 rajkumari startup.sh[32734]: Tomcat started.

```

30. How to print specific columns from a delimited file.

Objective: To display only selected columns from a file where fields are separated by a delimiter (such as comma, colon, or tab).

Purpose: This is useful for extracting only the required information from structured files like CSV files, log files, or /etc/passwd.

Using **cut** command

-d → Specifies the delimiter.

-f → Specifies the field (column) number(s)

```

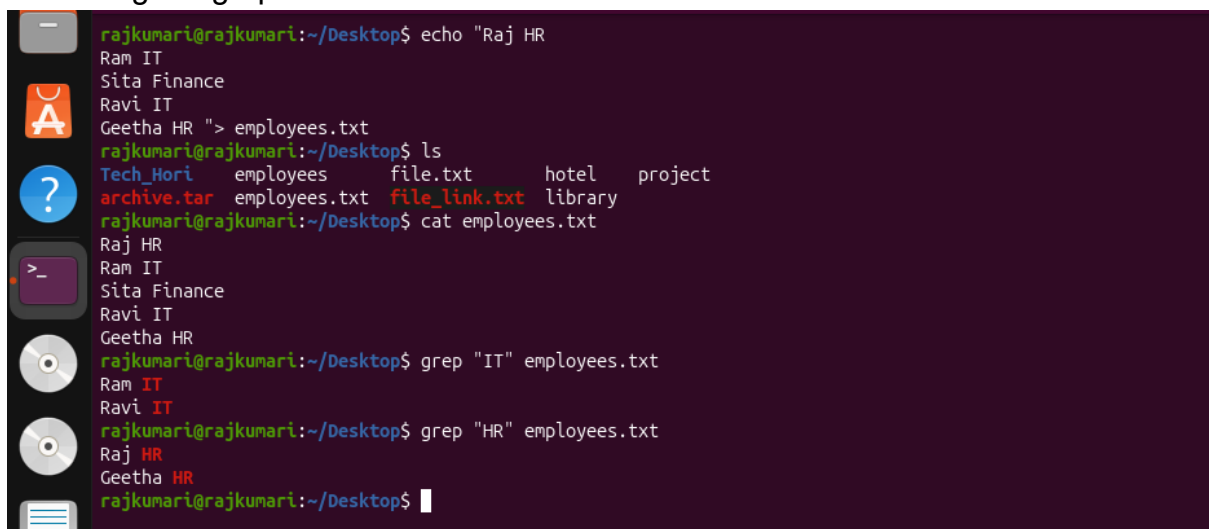
rajkumari@rajkumari:~/Desktop$ echo "101,Raj,HR
102,Ram,IT
103,Sita,Finance" > employees
rajkumari@rajkumari:~/Desktop$ ls
Tech_Hori  employees  file_link.txt  library
archive.tar file.txt    hotel          project
rajkumari@rajkumari:~/Desktop$ cat employees
101,Raj,HR
102,Ram,IT
103,Sita,Finance
rajkumari@rajkumari:~/Desktop$ cut -d ',' -f2 employees
Raj
Ram
Sita
rajkumari@rajkumari:~/Desktop$ cut -d ',' -f1,3 employees
101,HR
102,IT
103,Finance
rajkumari@rajkumari:~/Desktop$

```

31. How to filter and print lines based on a specific pattern or condition.

Objective: To search and display lines that match a specific pattern from a file.

Purpose: To quickly find required information from text files, logs, or configuration files using the grep command.



```

rajkumari@rajkumari:~/Desktop$ echo "Raj HR
Ram IT
Sita Finance
Ravi IT
Geetha HR" > employees.txt
rajkumari@rajkumari:~/Desktop$ ls
Tech_Hori  employees  file.txt    hotel  project
archive.tar employees.txt file_link.txt library
rajkumari@rajkumari:~/Desktop$ cat employees.txt
Raj HR
Ram IT
Sita Finance
Ravi IT
Geetha HR
rajkumari@rajkumari:~/Desktop$ grep "IT" employees.txt
Ram IT
Ravi IT
rajkumari@rajkumari:~/Desktop$ grep "HR" employees.txt
Raj HR
Geetha HR
rajkumari@rajkumari:~/Desktop$

```

32. How to calculate and print the average, sum, or other statistics of a column.

Objective: To calculate the **sum, average, count, minimum, or maximum** values of a numeric column in a file.

Purpose: To analyze numeric data from text files using the awk command.

```
rajkumari@rajkumari:~/Desktop$ echo "Raj 80
Raj 70
Sita 90
Ravi 60
Geetha 100" > marks.txt
rajkumari@rajkumari:~/Desktop$ cat marks.txt
Raj 80
Raj 70
Sita 90
Ravi 60
Geetha 100
rajkumari@rajkumari:~/Desktop$ awk '{sum += $2} END {print "Sum =", sum}' marks.txt
Sum = 400
rajkumari@rajkumari:~/Desktop$ awk '{sum += $2} END {print "Average =", sum/NR}' marks.txt
Average = 80
rajkumari@rajkumari:~/Desktop$ awk 'END {print "Count =", NR}' marks.txt
Count = 5
rajkumari@rajkumari:~/Desktop$
```

33. How to perform string manipulation, such as extracting substrings or changing case.

Objective: To manipulate strings in Linux by extracting substrings and converting text to uppercase or lowercase using commands like cut, awk, tr, and sed.

Purpose: Extract specific characters or fields from a string.

- Convert text between uppercase and lowercase.
- Modify strings for text processing and scripting.

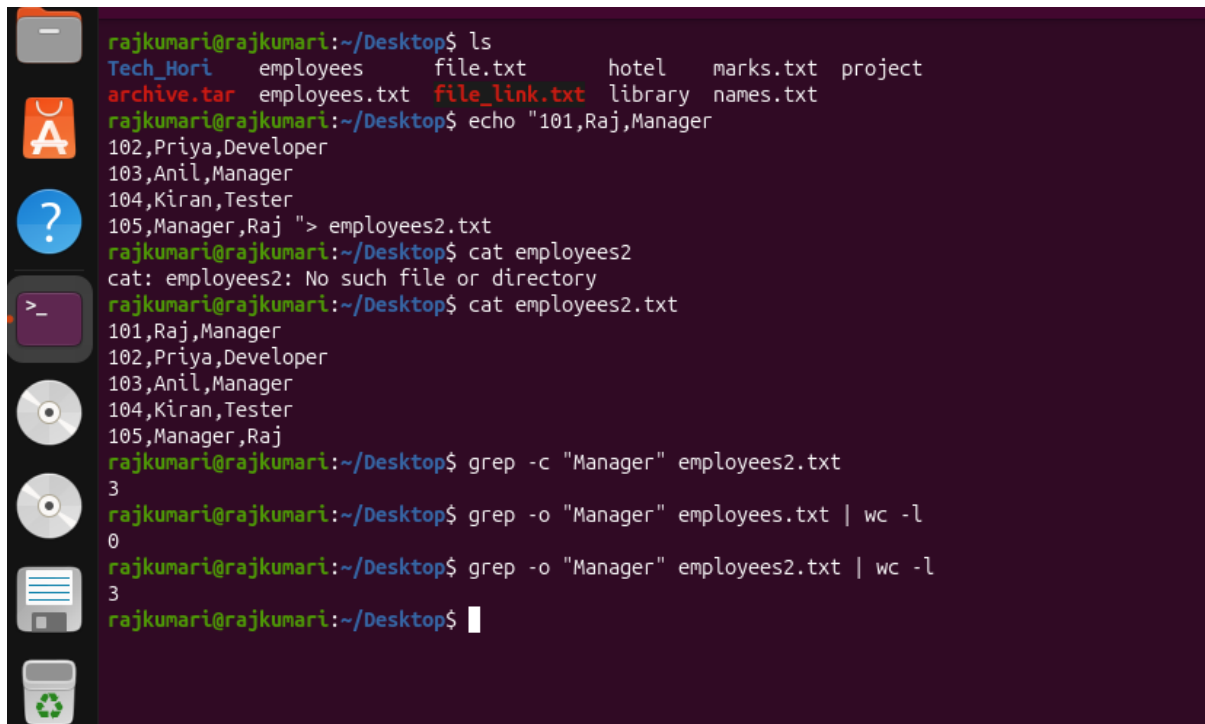
```
rajkumari@rajkumari:~/Desktop$ echo "Rajkumari
Linux
ApacheTomcat
DevOps" > names.txt
rajkumari@rajkumari:~/Desktop$ ls
Tech_Hori  employees  file.txt  hotel  marks.txt  project
archive.tar employees.txt file_link.txt library  names.txt
rajkumari@rajkumari:~/Desktop$ cat names.txt
Rajkumari
Linux
ApacheTomcat
DevOps
rajkumari@rajkumari:~/Desktop$ cut -c1-3 names.txt
Raj
Lin
Apa
Dev
rajkumari@rajkumari:~/Desktop$ cut -c4-8 names.txt
kumar
ux
cheTo
Ops
rajkumari@rajkumari:~/Desktop$ echo "linux administration" | tr 'a-z' 'A-Z'
LINUX ADMINISTRATION
rajkumari@rajkumari:~/Desktop$ echo "Rajkumari" | awk '{print substr($1,4,5)}'
kumar
rajkumari@rajkumari:~/Desktop$
```

34. How to count the occurrences of a specific pattern in a file.

The **grep -c** is used to count how many lines in a file contain a specific pattern. It is useful for quickly finding the number of matching entries in log files, configuration files, or text documents.

Objective: To count the number of occurrences of a specific pattern in a file using Linux commands.

Purpose: To quickly identify how many times a word, text, or pattern appears in a file. This is useful for analyzing log files, verifying data, and searching text efficiently.

A terminal window with a dark purple background and a sidebar on the left containing icons for file manager, applications, help, terminal, and other system utilities. The terminal shows the following commands and output:

```
rajkumari@rajkumari:~/Desktop$ ls
Tech_Hori  employees  file.txt  hotel  marks.txt  project
archive.tar employees.txt file_link.txt library names.txt
rajkumari@rajkumari:~/Desktop$ echo "101,Raj,Manager"
101,Raj,Manager
rajkumari@rajkumari:~/Desktop$ cat employees2
cat: employees2: No such file or directory
rajkumari@rajkumari:~/Desktop$ cat employees2.txt
101,Raj,Manager
102,Priya,Developer
103,Anil,Manager
104,Kiran,Tester
105,Manager,Raj
rajkumari@rajkumari:~/Desktop$ grep -c "Manager" employees2.txt
3
rajkumari@rajkumari:~/Desktop$ grep -o "Manager" employees.txt | wc -l
0
rajkumari@rajkumari:~/Desktop$ grep -o "Manager" employees2.txt | wc -l
3
rajkumari@rajkumari:~/Desktop$
```

35. How to sort lines based on a specific field or column.

sort → Sorts the lines in a file.

-t ' , ' → Uses a comma (,) as the field delimiter.

-k1 → Sorts by the **1st column**.

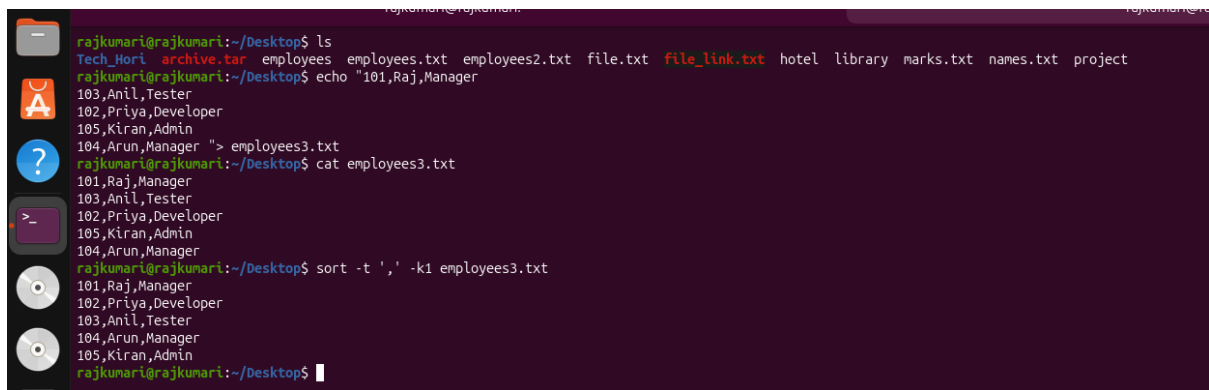
-k2 → Sorts by the **2nd column**.

-k3 → Sorts by the **3rd column**.

The **sort** command is used to arrange lines in a file. The -t option specifies the field delimiter, and the -k option specifies the field (column) to sort by.

Objective: To sort the contents of a file based on a specific field or column.

Purpose: To organize data in ascending or descending order based on a chosen field, making it easier to analyze and manage records.

A terminal window with a dark purple background and light green text. The prompt is 'rajkumari@rajkumari:~/Desktop\$'. The user runs 'ls', showing a list of files including 'Tech_Hori', 'archive.tar', 'employees', 'employees.txt', 'employees2.txt', 'file.txt', 'file_link.txt', 'hotel', 'library', 'marks.txt', 'names.txt', and 'project'. Then, they run 'echo "101,Raj,Manager"', which outputs '101,Raj,Manager'. Next, they run 'cat employees3.txt', which outputs '101,Raj,Manager', '103,Anil,Tester', '102,Priya,Developer', '105,Kiran,Admin', and '104,Arun,Manager'. Finally, they run 'sort -t ',' -k1 employees3.txt', which outputs the same five lines but sorted by the first column (ID), resulting in '101,Raj,Manager', '102,Priya,Developer', '103,Anil,Tester', '104,Arun,Manager', and '105,Kiran,Admin'.

```
rajkumari@rajkumari:~/Desktop$ ls
Tech_Hori archive.tar employees employees.txt employees2.txt file.txt file_link.txt hotel library marks.txt names.txt project
rajkumari@rajkumari:~/Desktop$ echo "101,Raj,Manager"
101,Raj,Manager
rajkumari@rajkumari:~/Desktop$ cat employees3.txt
101,Raj,Manager
103,Anil,Tester
102,Priya,Developer
105,Kiran,Admin
104,Arun,Manager
rajkumari@rajkumari:~/Desktop$ sort -t ',' -k1 employees3.txt
101,Raj,Manager
102,Priya,Developer
103,Anil,Tester
104,Arun,Manager
105,Kiran,Admin
rajkumari@rajkumari:~/Desktop$
```

36. How to merge multiple files based on a common field or column.

The **join** command merges two sorted files by comparing a common field (column). Both files must be sorted on the join field before using the **join** command.

Explanation:

join → Merges two files based on a common field.

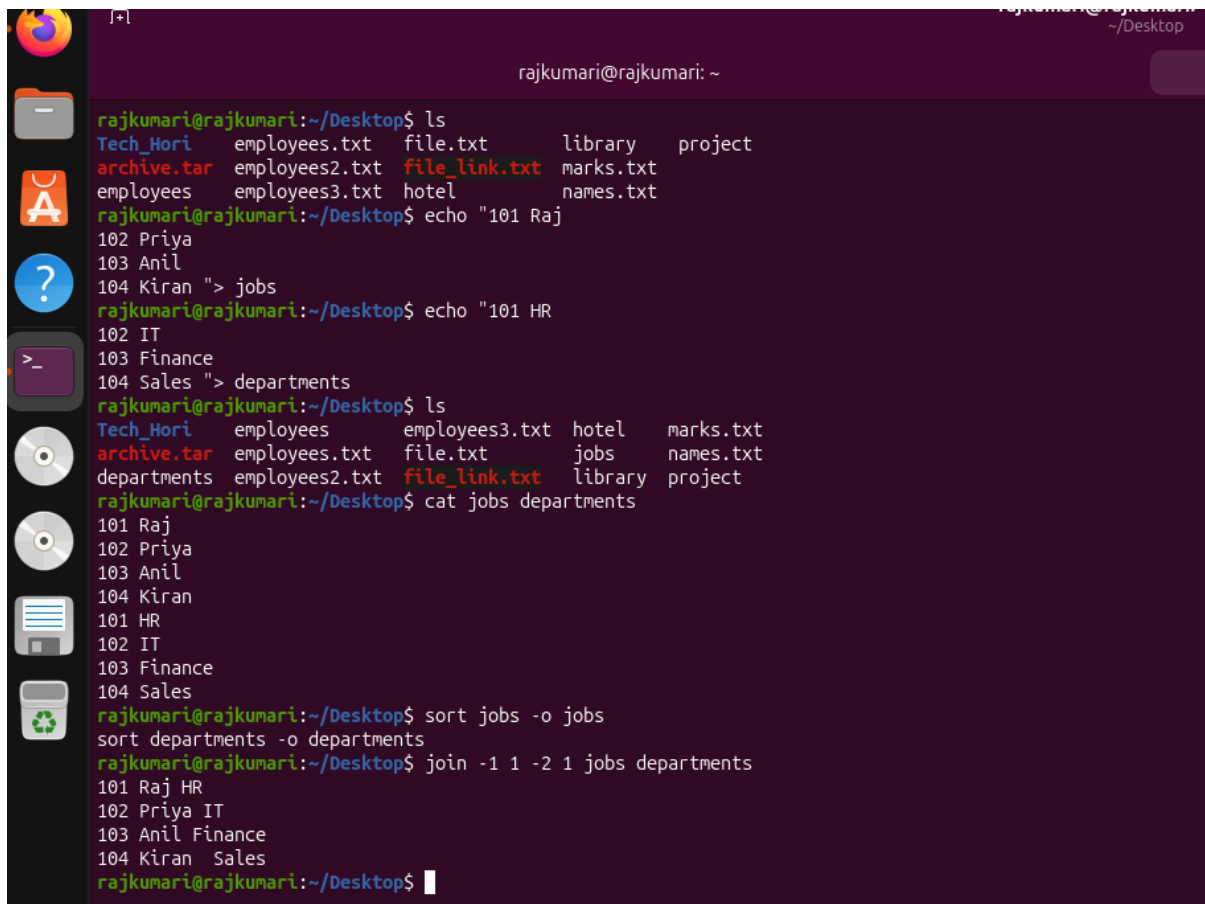
-1 1 → Uses the **1st column** of the first file as the join field.

-2 1 → Uses the **1st column** of the second file as the join field.

Both files **must be sorted** on the join column before running join.

Objective: To merge two files based on a common field or column using the join command.

Purpose: To combine related data from multiple files by matching a common field, making it easier to create a unified dataset.



```
rajkumari@rajkumari: ~/Desktop
rajkumari@rajkumari:~/Desktop$ ls
Tech_Hori  employees.txt  file.txt      library  project
archive.tar employees2.txt  file_link.txt marks.txt
employees  employees3.txt hotel         names.txt
rajkumari@rajkumari:~/Desktop$ echo "101 Raj
102 Priya
103 Anil
104 Kiran "> jobs
rajkumari@rajkumari:~/Desktop$ echo "101 HR
102 IT
103 Finance
104 Sales "> departments
rajkumari@rajkumari:~/Desktop$ ls
Tech_Hori  employees  employees3.txt  hotel  marks.txt
archive.tar employees.txt  file.txt      jobs  names.txt
departments employees2.txt  file_link.txt  library project
rajkumari@rajkumari:~/Desktop$ cat jobs departments
101 Raj
102 Priya
103 Anil
104 Kiran
101 HR
102 IT
103 Finance
104 Sales
rajkumari@rajkumari:~/Desktop$ sort jobs -o jobs
sort departments -o departments
rajkumari@rajkumari:~/Desktop$ join -1 1 -2 1 jobs departments
101 Raj HR
102 Priya IT
103 Anil Finance
104 Kiran Sales
rajkumari@rajkumari:~/Desktop$
```

37. How to substitute text in a file using search and replace

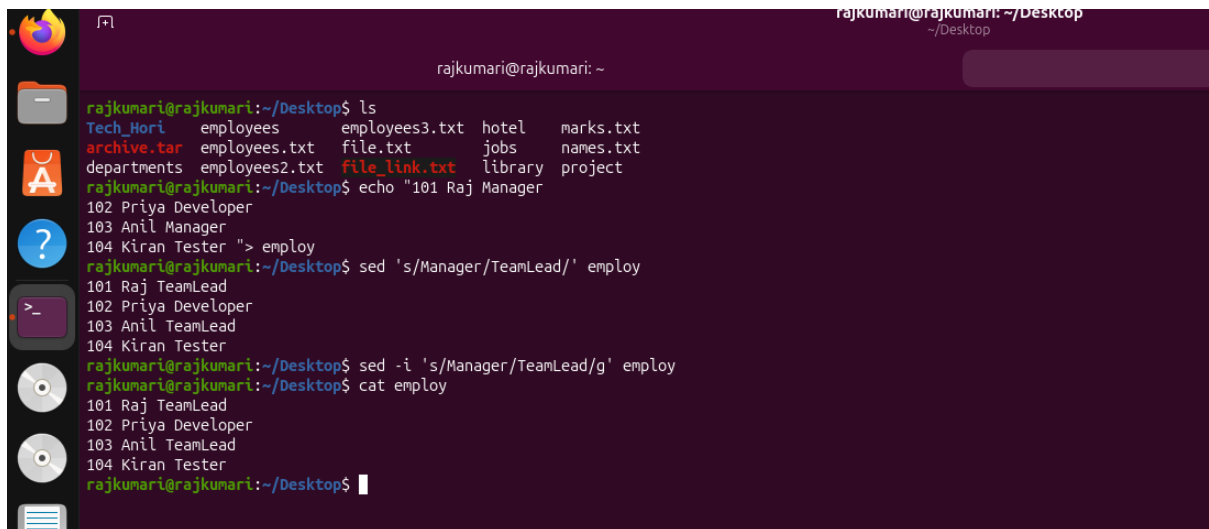
The **sed** (Stream Editor) command is used to search for a specific word or pattern and replace it with another word. It can display the modified output or update the file directly using the **-i** option.

Objective: To replace specific text in a file using the **sed** command.

Purpose: To update or modify text in a file efficiently without editing it manually.

Explanation:

- **sed** → Stream editor used for text processing.
- **s/old/new/** → Substitutes the first occurrence of old with new in each line.
- **g** → Replaces all occurrences of the text in each line.
- **-i** → Modifies the original file in place.

A terminal window with a dark purple background. The prompt is 'rajkumari@rajkumari: ~/Desktop'. The user runs 'ls', showing a directory listing of files like 'Tech_Hori', 'archive.tar', 'employees', etc. Then they run 'echo "101 Raj Manager"', adding a new line. Next, they run 'sed 's/Manager/TeamLead/' employ', which updates the line in place. Finally, they run 'cat employ', showing the updated content of the file.

```
rajkumari@rajkumari: ~/Desktop
rajkumari@rajkumari:~$ ls
Tech_Hori  employees      employees3.txt  hotel  marks.txt
archive.tar employees.txt  file.txt       jobs   names.txt
departments employees2.txt file_link.txt   library project
rajkumari@rajkumari:~$ echo "101 Raj Manager"
101 Raj Manager
rajkumari@rajkumari:~$ sed 's/Manager/TeamLead/' employ
101 Raj TeamLead
102 Priya Developer
103 Anil Manager
104 Kiran Tester "> employ
rajkumari@rajkumari:~$ sed -i 's/Manager/TeamLead/g' employ
rajkumari@rajkumari:~$ cat employ
101 Raj TeamLead
102 Priya Developer
103 Anil TeamLead
104 Kiran Tester
rajkumari@rajkumari:~$
```

38. How to Delete specific lines based on a pattern or line number.

The **sed** command is used to delete lines from a file. You can delete a line by specifying its line number or by matching a specific pattern.

Explanation

- **sed '3d' file** → Deletes the **3rd line** (display only).
- **sed -i '3d' file** → Deletes the **3rd line permanently**.
- **sed '/pattern/d' file** → Deletes all lines containing the specified **pattern** (display only).
- **sed -i '/pattern/d' file** → Deletes all matching lines **permanently**.

```
rajkumari@rajkumari:~/Desktop$ ls
Tech_Hori    employ      employees2.txt  file_link.txt  library    project
archive.tar  employees   employees3.txt  hotel          marks.txt
departments  employees.txt file.txt        jobs           names.txt
rajkumari@rajkumari:~/Desktop$ echo "101 Raj Manager
102 Priya Developer
103 Anil Manager
104 Kiran Tester
105 Arun Admin "> holder
rajkumari@rajkumari:~/Desktop$ cat holder
101 Raj Manager
102 Priya Developer
103 Anil Manager
104 Kiran Tester
105 Arun Admin
rajkumari@rajkumari:~/Desktop$ sed '3d' holder
101 Raj Manager
102 Priya Developer
104 Kiran Tester
105 Arun Admin
rajkumari@rajkumari:~/Desktop$ sed -i '3d' holder
rajkumari@rajkumari:~/Desktop$ cat holder
101 Raj Manager
102 Priya Developer
104 Kiran Tester
105 Arun Admin
rajkumari@rajkumari:~/Desktop$ sed '/Manager/d' holder
102 Priya Developer
104 Kiran Tester
105 Arun Admin
rajkumari@rajkumari:~/Desktop$ sed -i '/Manager/d' holder
rajkumari@rajkumari:~/Desktop$ cat holder
102 Priya Developer
104 Kiran Tester
105 Arun Admin
rajkumari@rajkumari:~/Desktop$
```

39. How to append or insert text before or after a specific pattern or line.

The **sed** command is used to insert or append text in a file. The **i** option inserts text **before** a line or pattern, while the option appends text **after** a line or pattern.

Explanation

- **i** → Inserts text **before** a specified line or matching pattern.
- **a** → Appends text **after** a specified line or matching pattern.
- **-i** → Saves the changes permanently in the file.

Objective: To insert or append text before or after a specific line or pattern in a file.

Purpose: To add new information to a file at a desired location without manually editing the file.

```
rajkumari@rajkumari:~/Desktop$ ls
Tech_Hori  employees  file.txt      jobs      project
archive.tar employees.txt file_link.txt library
departments employees2.txt holder      marks.txt
employ     employees3.txt hotel       names.txt
rajkumari@rajkumari:~/Desktop$ echo "101 Raj Manager"
102 Priya Developer
103 Anil Tester
104 Kiran Admin "> teachers
rajkumari@rajkumari:~/Desktop$ cat teachers
101 Raj Manager
102 Priya Developer
103 Anil Tester
104 Kiran Admin
rajkumari@rajkumari:~/Desktop$ sed '1i\100 Ravi Intern' teachers
100 Ravi Intern
101 Raj Manager
102 Priya Developer
103 Anil Tester
104 Kiran Admin
rajkumari@rajkumari:~/Desktop$ sed -i '1i\100 Ravi Intern' teachers
rajkumari@rajkumari:~/Desktop$ sed '2a\105 Arun TeamLead' teachers
100 Ravi Intern
101 Raj Manager
105 Arun TeamLead
102 Priya Developer
103 Anil Tester
104 Kiran Admin
rajkumari@rajkumari:~/Desktop$ sed -i '2a\105 Arun TeamLead' teachers
rajkumari@rajkumari:~/Desktop$ cat teachers
100 Ravi Intern
101 Raj Manager
105 Arun TeamLead
102 Priya Developer
103 Anil Tester
104 Kiran Admin
rajkumari@rajkumari:~/Desktop$
```

40. How to print only specific lines from a file.

The **sed** command is commonly used to print specific lines from a file. You can specify a single line number or a range of line numbers.

Explanation

- **sed** → Stream editor used for text processing.
- **-n** → Suppresses the default output.
- **'3p'** → Prints only the **3rd line**.
- **'2,4p'** → Prints lines **2 through 4**.
- **'\$p'** → Prints the **last line** of the file.

Objective: To display only the required line(s) from a file.

Purpose: To view specific lines without displaying the entire file, making it easier to locate and inspect particular data.

- **awk** is mainly used for data extraction, field processing, filtering, and calculations.
- **sed** is mainly used for editing text, such as searching, replacing, inserting, deleting, and printing lines in files.

```
rajkumari@rajkumari:~/Desktop$ ls
Tech_Hori    employees    file.txt     jobs         project
archive.tar  employees.txt file_link.txt library       teachers
departments employees2.txt holder       marks.txt
employ       employees3.txt hotel        names.txt
rajkumari@rajkumari:~/Desktop$ echo "101 Raj Manager
102 Priya Developer
103 Anil Tester
104 Kiran Admin
105 Arun TeamLead "> doctor.txt
rajkumari@rajkumari:~/Desktop$ cat doctor
cat: doctor: No such file or directory
rajkumari@rajkumari:~/Desktop$ cat doctor.txt
101 Raj Manager
102 Priya Developer
103 Anil Tester
104 Kiran Admin
105 Arun TeamLead
rajkumari@rajkumari:~/Desktop$ sed -n '3p' doctor.txt
103 Anil Tester
rajkumari@rajkumari:~/Desktop$ sed -n '2,4p' doctor.txt
102 Priya Developer
103 Anil Tester
104 Kiran Admin
rajkumari@rajkumari:~/Desktop$ sed -n '1p' doctor.txt
101 Raj Manager
rajkumari@rajkumari:~/Desktop$ sed -n '$p' doctor.txt
105 Arun TeamLead
rajkumari@rajkumari:~/Desktop$
```

41. 5 use cases for AWK and 5 use cases for sed

Use Case of awk	Description	Example Command
1. Print specific columns	Extract selected columns from a file.	awk '{print \$1, \$3}' employees.txt
2. Filter records	Print lines matching a condition or pattern.	awk '\$3=="Manager"' employees.txt
3. Perform calculations	Calculate sum, average, minimum, or maximum values.	awk '{sum+=\$2} END {print sum}' marks.txt
4. Count records	Count the number of lines or matching records.	awk 'END {print NR}' employees.txt
5. Change field separator	Process files with delimiters such as commas or colons.	awk -F',' '{print \$2}' employees.csv

Use Case of sed	Description	Example Command
1. Search and replace text	Replace text in a file.	sed 's/Manager/TeamLead/g' employees.txt
2. Delete specific lines	Remove lines by line number or pattern.	sed '3d' employees.txt
3. Insert or append text	Add text before or after a line or pattern.	sed '2i\New Employee' employees.txt
4. Print specific lines	Display only selected lines from a file.	sed -n '2,4p' employees.txt
5. Replace text permanently	Modify the original file directly.	sed -i 's/Developer/Engineer/g' employees.txt

42. How to copy file from linux to windows machine.

Objective: To copy a file from an Ubuntu virtual machine to a Windows machine using VMware Shared Folders.

Purpose: To securely transfer files between Ubuntu and Windows without using USB drives or network file transfer.

Step 1: Create a Shared Folder in Windows

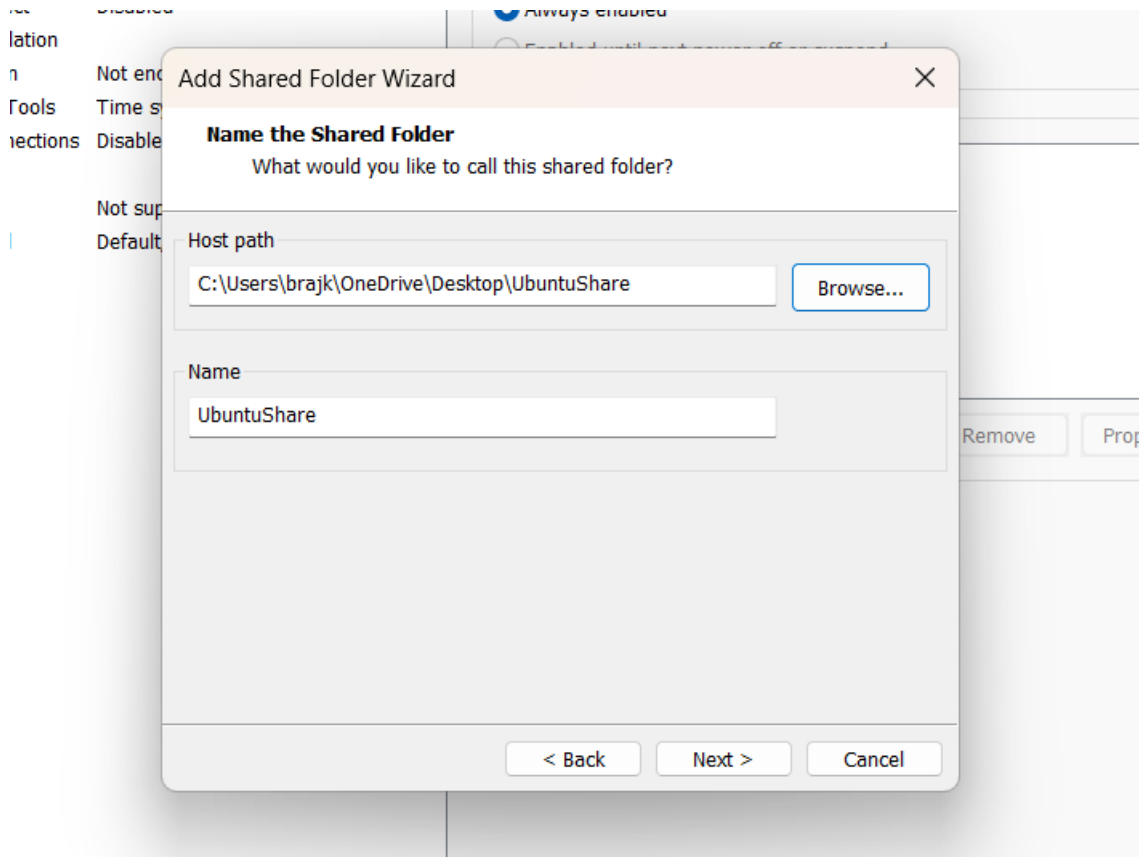
1. On Windows Desktop, right-click.
2. Select New → Folder.
3. Name the folder: UbuntuShare

Example location: C:\Users\brajk\OneDrive\Desktop\UbuntuShare

Step 2: Configure VMware Shared Folder

1. Shut down the Ubuntu VM.
2. Open **VMware Workstation**.
3. Select the Ubuntu VM.
4. Go to: VM → Settings
5. Click the Options tab.
6. Select Shared Folder
7. Choose: Always enabled
8. Click: Add
9. Browse to: C:\Users\brajk\OneDrive\Desktop\UbuntuShare

10. Click next
11. Keep the share Name as: UbuntuShare
12. Click: Finish → OK



Step 3: Start Ubuntu

Power on the Ubuntu virtual machine.

Step 4: Verify the Shared Folder

Run: `$ vmware-hgfsclient`

Step 5: Create the Mount Point

Run: `$ sudo mkdir -p /mnt/hgfs`

Step 6: Mount the Shared Folder

Run: `$ sudo vmhgfs-fuse .host:/ /mnt/hgfs -o allow_other`

Step 7: Verify the Mount

Run: `ls /mnt/hgfs`

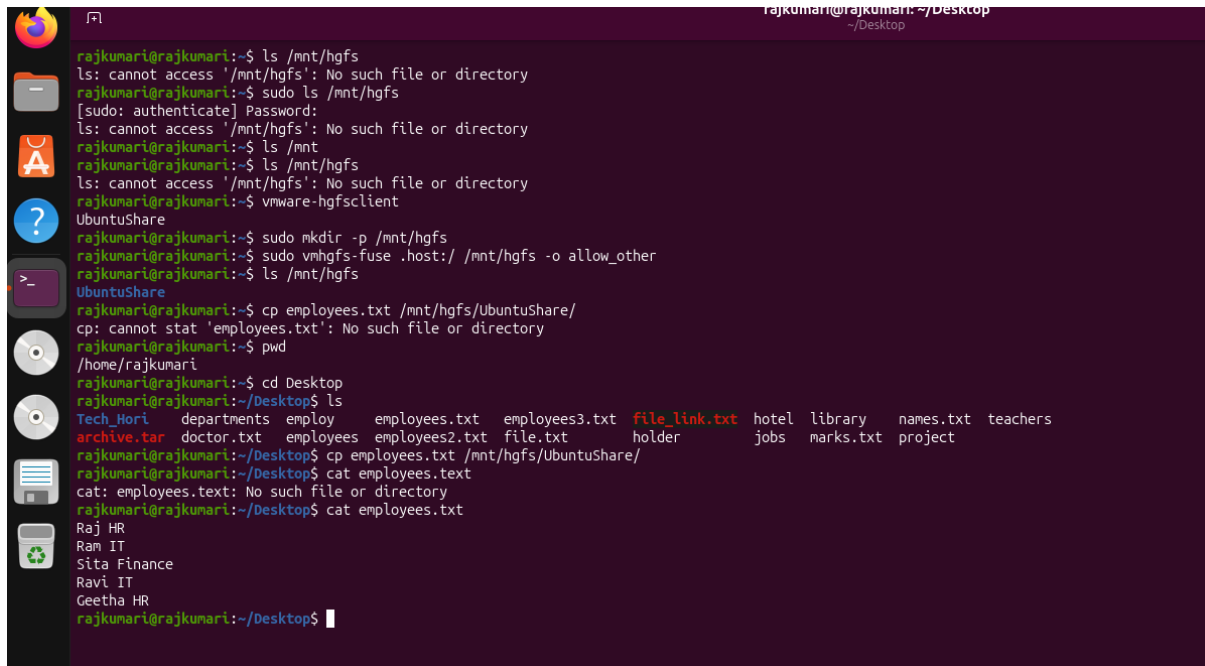
Step 8: Go to the File Location

Example: `$ cd /home/rajkumari/Desktop`

Or Run: `$ pwd`

Step 9: Copy the File

If the file is in the current directory: `$ cp employees.txt /mnt/hgfs/UbuntuShare/`

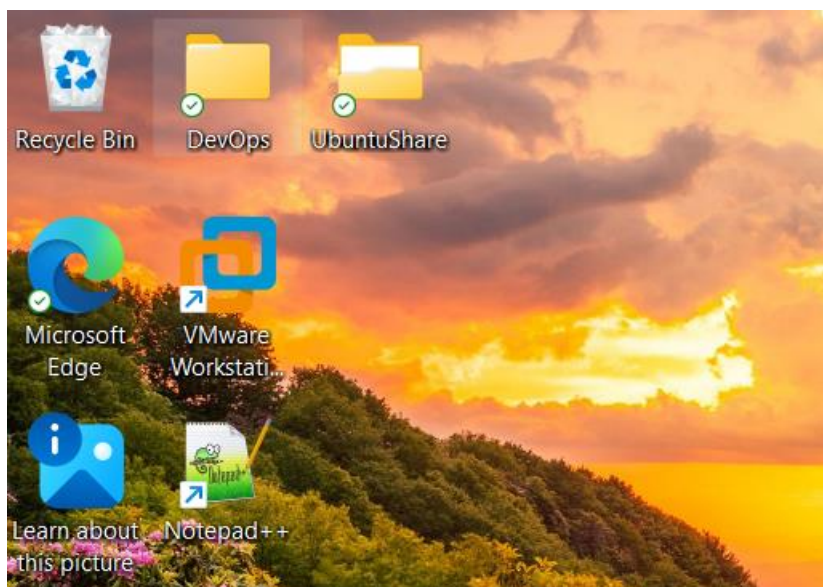


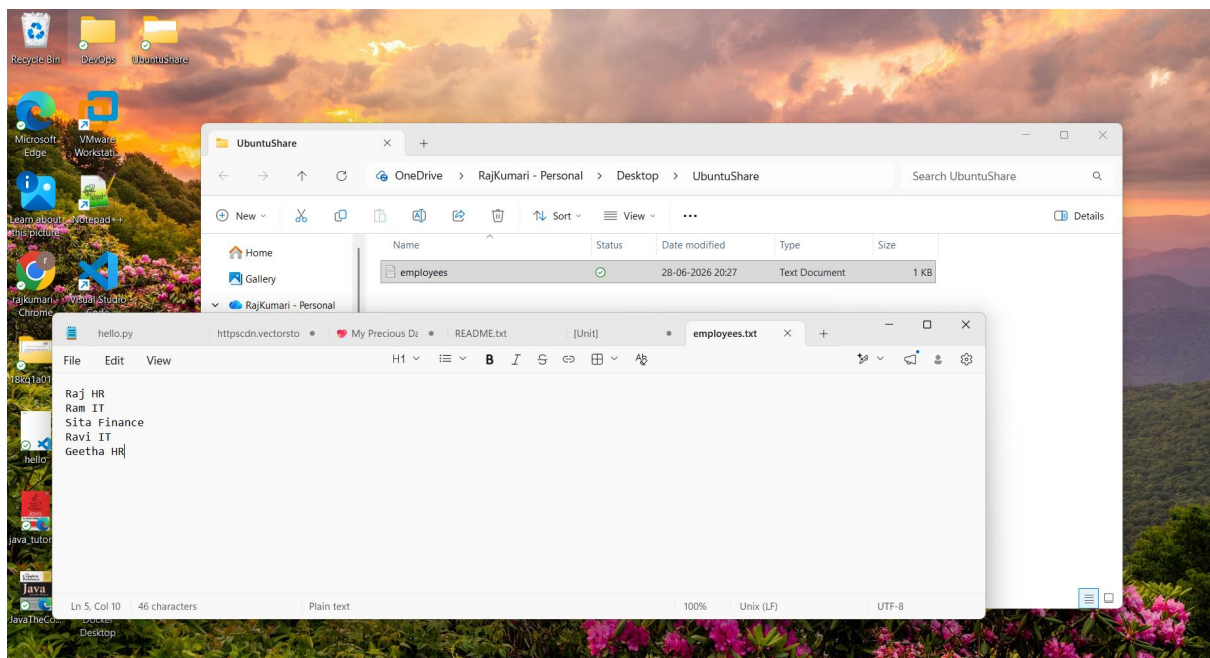
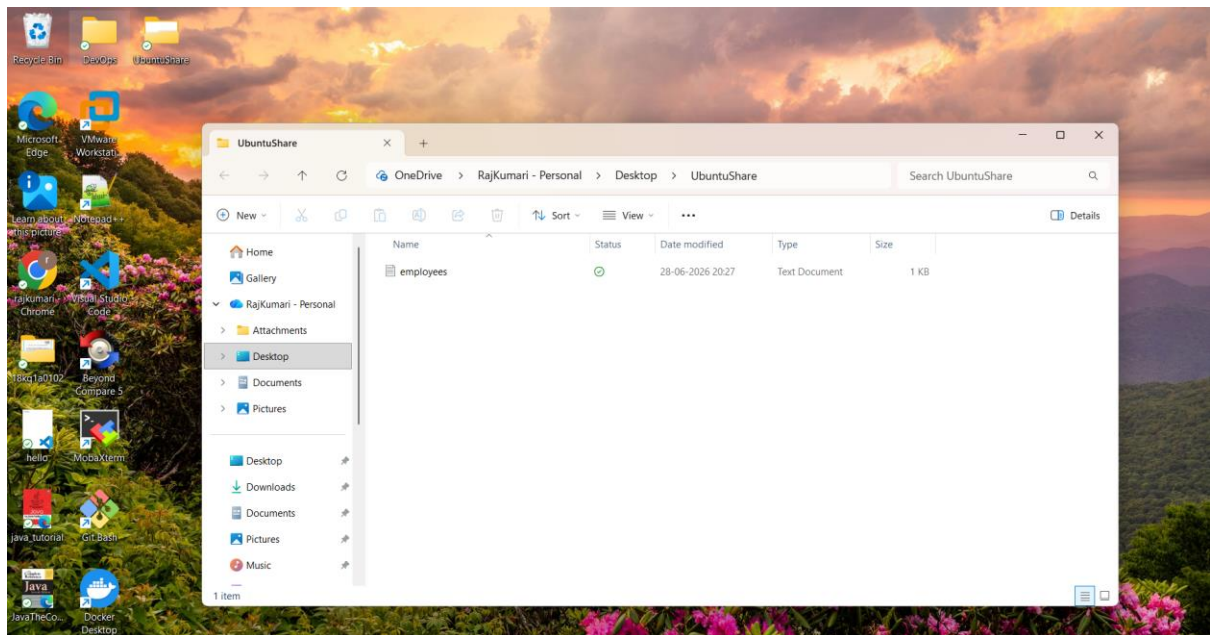
```
rajkumari@rajkumari:~$ ls /mnt/hgfs
ls: cannot access '/mnt/hgfs': No such file or directory
rajkumari@rajkumari:~$ sudo ls /mnt/hgfs
[sudo: authenticate] Password:
ls: cannot access '/mnt/hgfs': No such file or directory
rajkumari@rajkumari:~$ ls /mnt
rajkumari@rajkumari:~$ ls /mnt/hgfs
ls: cannot access '/mnt/hgfs': No such file or directory
rajkumari@rajkumari:~$ vmware-hgfsclient
UbuntuShare
rajkumari@rajkumari:~$ sudo mkdir -p /mnt/hgfs
rajkumari@rajkumari:~$ sudo vmhgfs-fuse .host:/ /mnt/hgfs -o allow_other
rajkumari@rajkumari:~$ ls /mnt/hgfs
UbuntuShare
rajkumari@rajkumari:~$ cp employees.txt /mnt/hgfs/UbuntuShare/
cp: cannot stat 'employees.txt': No such file or directory
rajkumari@rajkumari:~$ pwd
/home/rajkumari
rajkumari@rajkumari:~$ cd Desktop
rajkumari@rajkumari:~/Desktop$ ls
Tech_Hori  departments  employ      employees.txt  employees3.txt  file.link.txt  hotel  library  names.txt  teachers
archive.tar  doctor.txt  employees  employees2.txt  file.txt        holder         jobs   marks.txt  project
rajkumari@rajkumari:~/Desktop$ cp employees.txt /mnt/hgfs/UbuntuShare/
rajkumari@rajkumari:~/Desktop$ cat employees.txt
cat: employees.txt: No such file or directory
rajkumari@rajkumari:~/Desktop$ cat employees.txt
Raj  HR
Ram  IT
Sita Finance
Ravi  IT
Geetha HR
rajkumari@rajkumari:~/Desktop$
```

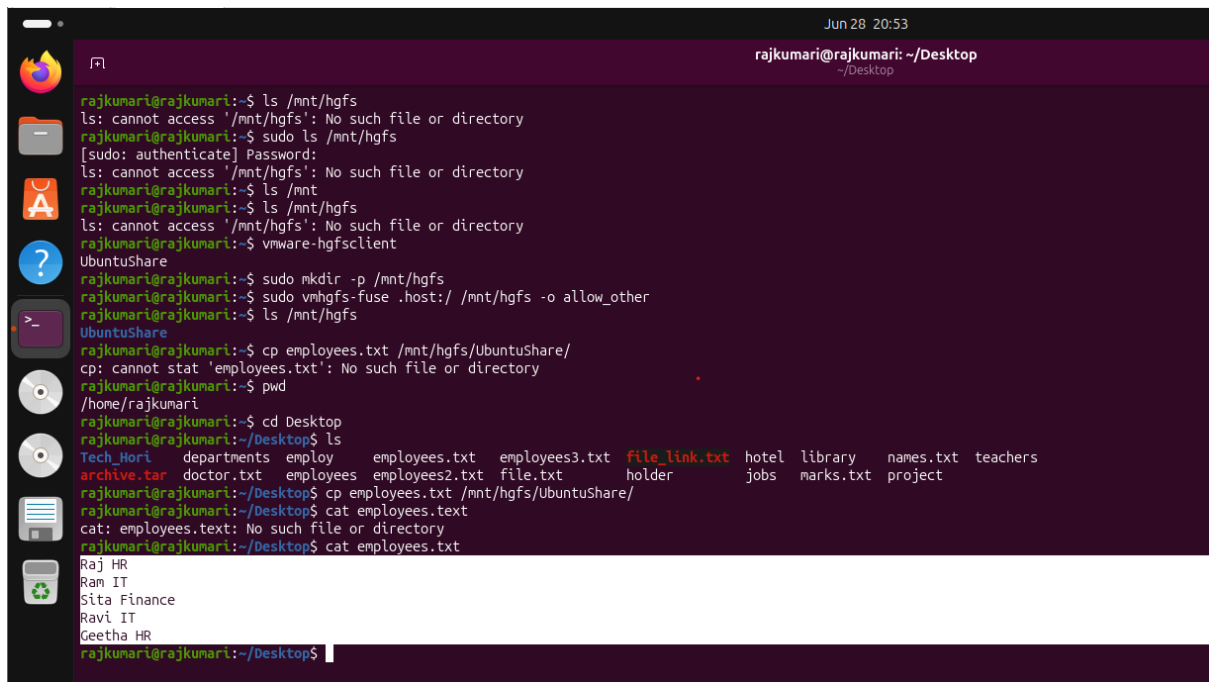
Step 10: Verify on Windows

Open: `C:\Users\brajk\OneDrive\Desktop\UbuntuShare`

You should see: **employees.txt**







A terminal window with a dark purple background. The title bar shows the date and time 'Jun 28 20:53' and the user and location 'rajkumari@rajkumari: ~/Desktop'. The terminal displays a series of commands and their outputs. The user attempts to list files in '/mnt/hgfs' but fails. They then try to list files in '/mnt' and use 'vmware-hgfsclient' to access 'UbuntuShare'. After creating the directory '/mnt/hgfs' and mounting it with 'vmhgfs-fuse', they attempt to copy 'employees.txt' from 'UbuntuShare' but fail. Finally, they navigate to the Desktop, list files, and successfully copy 'employees.txt' to the mounted share. The output of 'cat employees.txt' shows a list of names and roles.

```
rajkumari@rajkumari:~$ ls /mnt/hgfs
ls: cannot access '/mnt/hgfs': No such file or directory
rajkumari@rajkumari:~$ sudo ls /mnt/hgfs
[sudo: authenticate] Password:
ls: cannot access '/mnt/hgfs': No such file or directory
rajkumari@rajkumari:~$ ls /mnt
rajkumari@rajkumari:~$ ls /mnt/hgfs
ls: cannot access '/mnt/hgfs': No such file or directory
rajkumari@rajkumari:~$ vmware-hgfsclient
UbuntuShare
rajkumari@rajkumari:~$ sudo mkdir -p /mnt/hgfs
rajkumari@rajkumari:~$ sudo vmhgfs-fuse .host:/ /mnt/hgfs -o allow_other
rajkumari@rajkumari:~$ ls /mnt/hgfs
UbuntuShare
rajkumari@rajkumari:~$ cp employees.txt /mnt/hgfs/UbuntuShare/
cp: cannot stat 'employees.txt': No such file or directory
rajkumari@rajkumari:~$ pwd
/home/rajkumari
rajkumari@rajkumari:~$ cd Desktop
rajkumari@rajkumari:~/Desktop$ ls
Tech_Hori  departments  employ       employees3.txt  file_link.txt  hotel  library  names.txt  teachers
archive.tar  doctor.txt  employees   employees2.txt  file.txt       holder  jobs    marks.txt  project
rajkumari@rajkumari:~/Desktop$ cp employees.txt /mnt/hgfs/UbuntuShare/
rajkumari@rajkumari:~/Desktop$ cat employees.txt
cat: employees.txt: No such file or directory
rajkumari@rajkumari:~/Desktop$ cat employees.txt
Raj HR
Ran IT
Sita Finance
Ravi IT
Geetha HR
rajkumari@rajkumari:~/Desktop$
```

Finally, The file was successfully copied from the **Ubuntu remote machine (VMware Virtual Machine)** to the **Windows local machine (Host System)** using **VMware Shared Folders**. The copied file is now available in the shared folder on the Windows machine.

